Different renewable resources

Renewable energy

- Renewable energy is energy that comes from a source that won't run out.
- They are natural and self-replenishing, and usually have a low- or zero-carbon footprint.
- A carbon footprint is the total amount of greenhouse gases (including carbon dioxide and methane) that are generated by our actions.

Renewable energy

ADVANTAGES	**DISADVANTAGES**
Renewable energy won’t run out.	Renewable energy has high upfront costs.
Renewable energy has lower maintenance requirements.	Renewable energy is intermittent.
Renewables save money.	Renewables have limited storage capabilities.
Renewable energy has numerous environmental benefits.	Renewable energy sources have geographic limitations.
Renewables lower reliance on foreign energy sources.	Renewables aren’t always 100% carbon-free.
Renewable energy leads to cleaner water and air.	
Renewable energy creates jobs.	
Renewable energy can cut down on waste.	

Types of renewable energy

- Solar
  - Pv
  - Thermal
- Wind
- Geothermal
- Tidal
- Ocean Thermal
- Hydro
- Biomass
- Nuclear

Solar Energy

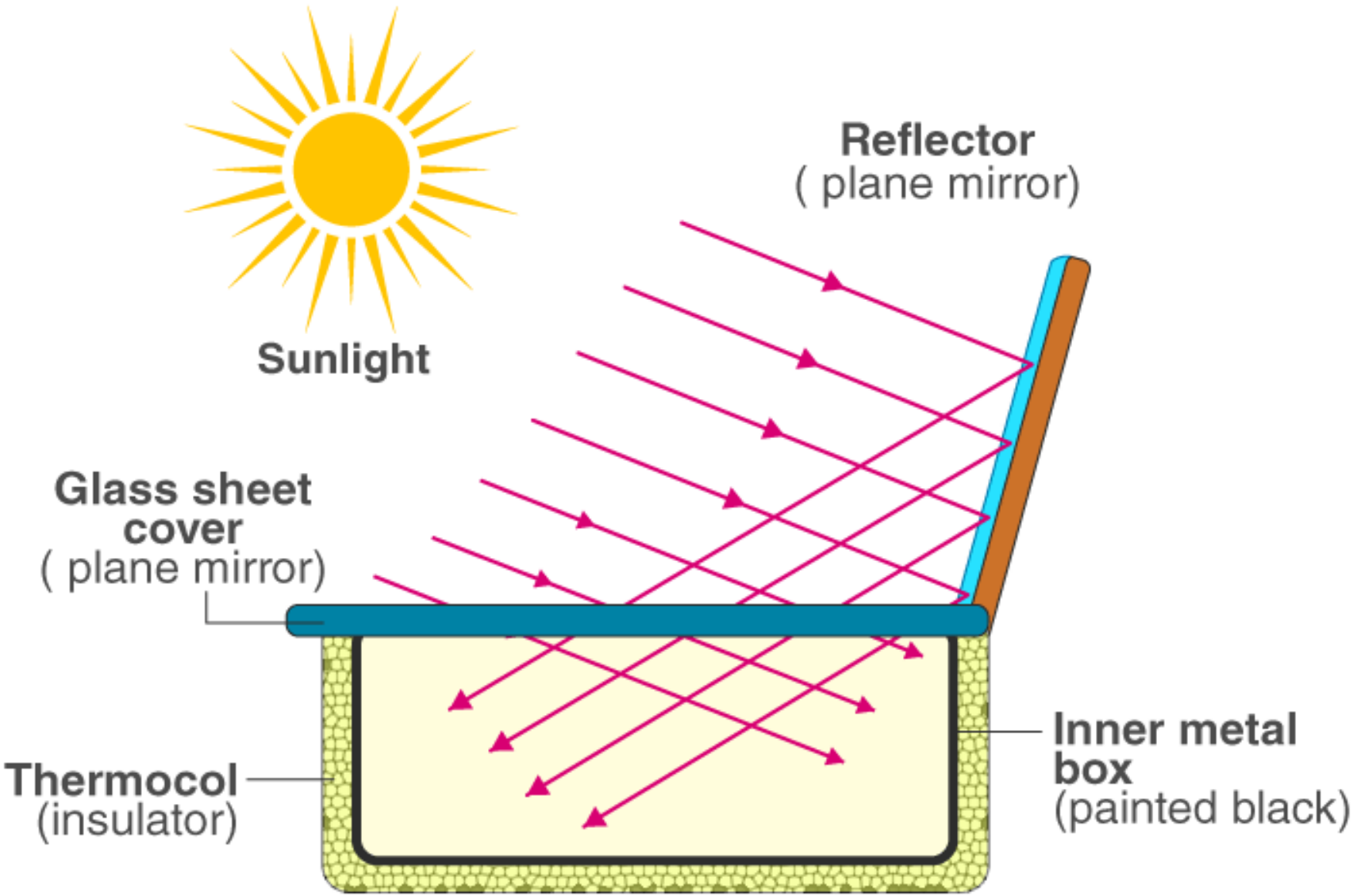
- Solar power is energy from the sun that is converted into thermal or electrical energy.
- Types of Solar energy
  - Direct
  - Indirect
- Direct solar energy
  - Solar heater
  - Solar pond
  - Solar cooker
- Indirect Solar energy
  - Solar PV
  - Solar light
  - Solar cells

Solar Energy

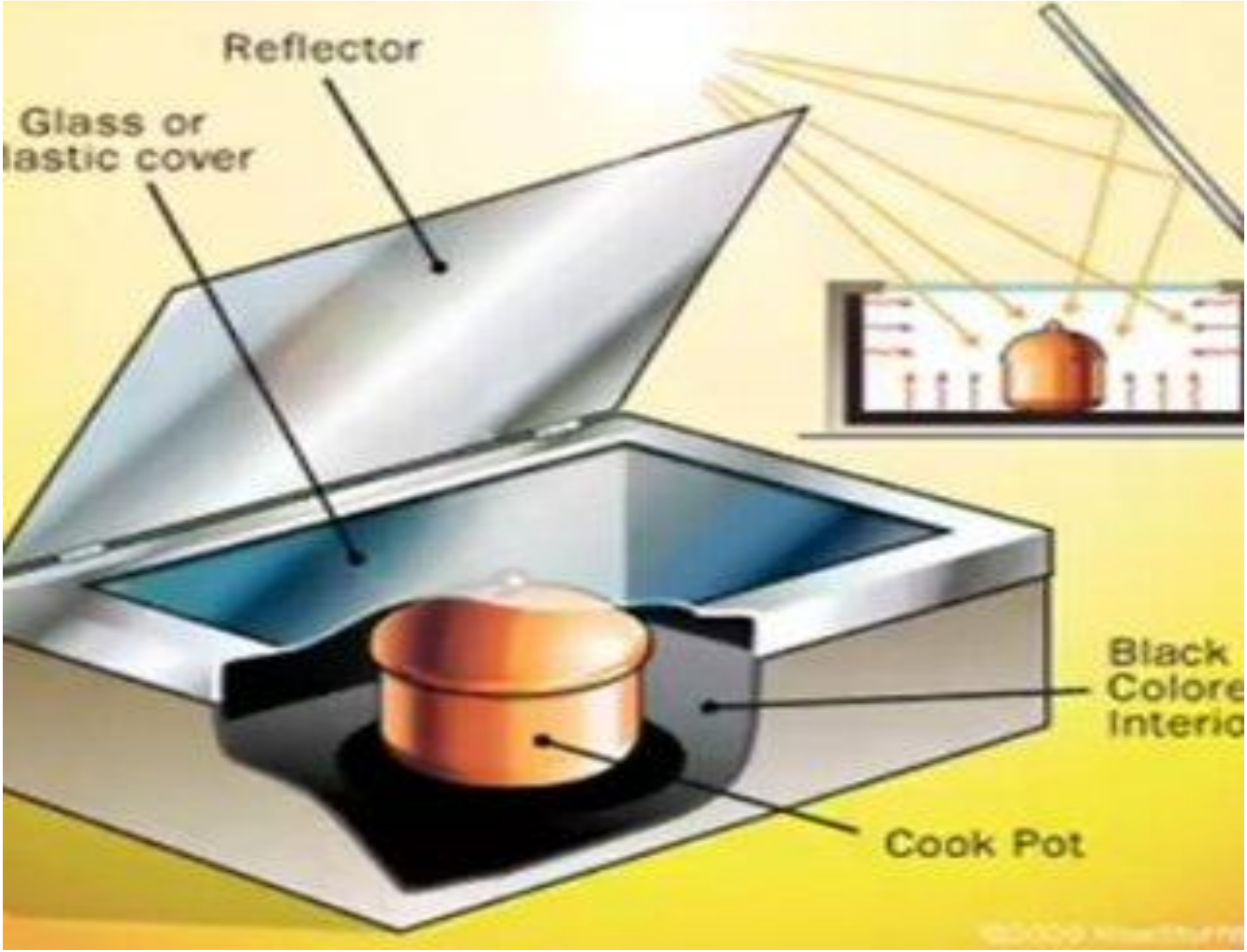
- **Direct solar energy:**
- Uses solar panels or collectors to directly absorb sunlight and convert it into usable energy.
- Examples: Solar photovoltaic panels generating electricity, flat-plate solar collectors for heating water.
- Requires direct exposure to sunlight for optimal efficiency.
- **Indirect solar energy:**
- Utilizes the effects of solar radiation on other systems to generate energy.
- Examples: Wind power (driven by solar-heated air currents), hydropower (from water evaporated by the sun), biomass energy (from plants using sunlight for photosynthesis).
- May not require direct sunlight, but depends on the secondary energy source created by the sun.

Solar cooker

- A solar cooker is a type of solar thermal collector.
- It “gathers” and traps the Sun's thermal (heat) energy.
- Heat is produced when high frequency light (visible and ultraviolet) is converted into low frequency infrared radiation.
- Solar cooker works on the principle to convert sunlight into heat energy and the heat energy trapped is used for cooking.

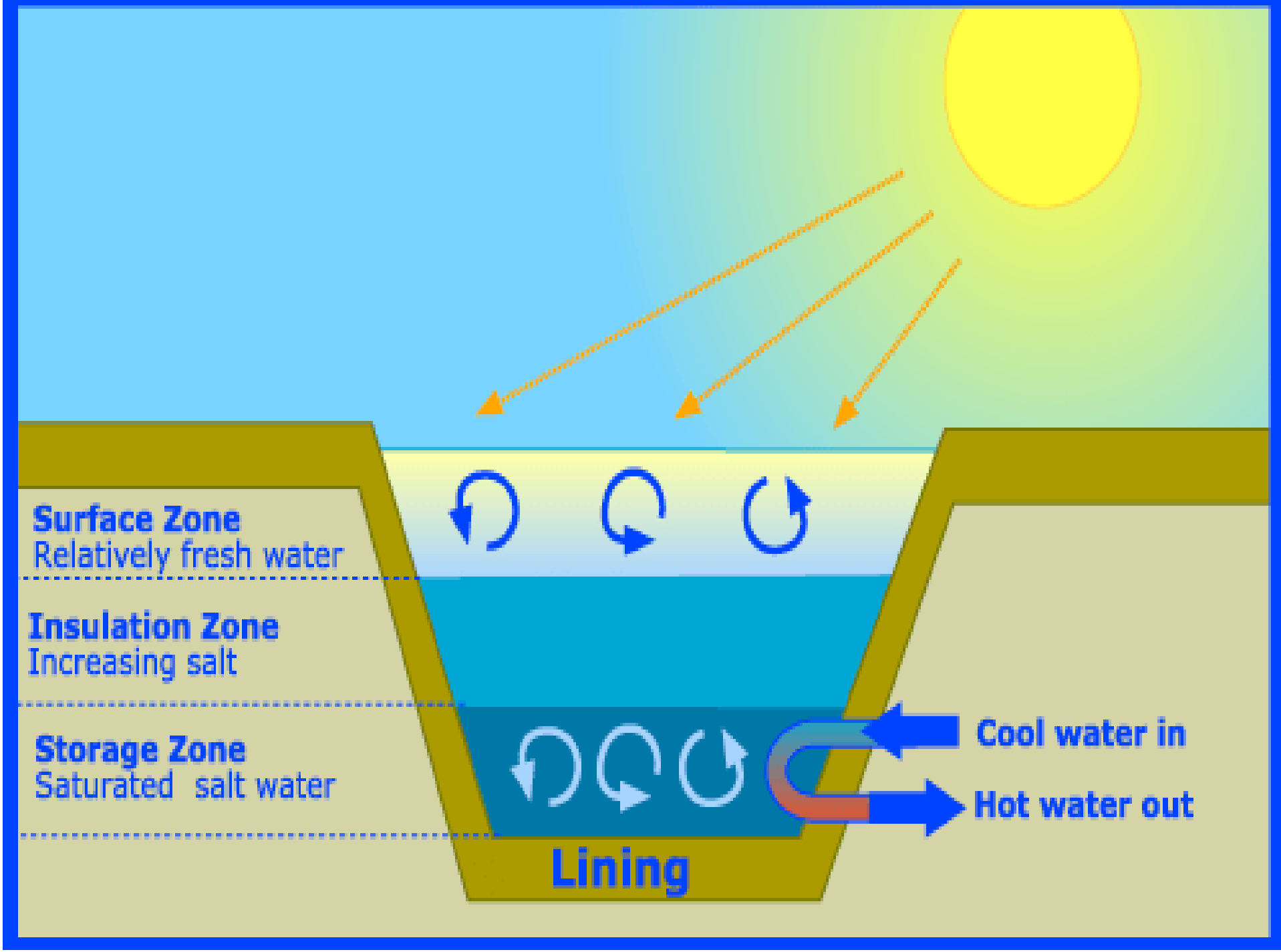


Solar Cooker


Reflector
Glass or
lastic cover
Black
Colore
Interio
Cook Pot

Solar pond

- A solar pond is a body of water that collects and stores solar energy.
- Solar energy will warm a body of water (that is exposed to the sun), but the water loses its heat unless some method is used to trap it.
- Water warmed by the sun expands and rises as it becomes less dense.
- Once it reaches the surface, the water loses its heat to the air through convection, or evaporates, taking heat with it.
- The colder water, which is heavier, moves down to replace the warm water, creating a natural convective circulation that mixes the water and dissipates the heat.
- The design of solar ponds reduces either convection or evaporation in order to store the heat collected by the pond. They can operate in almost any climate .
- A solar pond can store solar heat much more efficiently than a body of water of the same size because the salinity gradient prevents convection currents.
- Solar radiation entering the pond penetrates through to the lower layer, which contains concentrated salt solution.
- The temperature in this layer rises since the heat it absorbs from the sunlight is unable to move upwards to the surface by convection. Solar heat is thus stored in the lower layer of the pond .


Surface Zone
Relatively fresh water
Insulation Zone
Increasing salt
Storage Zone
Saturated salt water
Cool water in
Hot water out
Lining

Application

- Salt production (for enhanced evaporation or purification of salt, that is production of ‘vacuum quality’ salt)
- Aquaculture, using saline or fresh water (to grow, for example, fishor brine shrimp)
- Dairy industry (for example, to preheat feed water to boilers)
- Fruit and vegetable canning industry
- Fruit and vegetable drying (for example, vine fruit drying)
- Grain industry (for grain drying)
- Water supply (for desalination).

Solar heater
Solar Water Heater

- Solar water heating system is a device that helps in heating water by using the energy from the SUN.
- Solar energy (sun rays) is used for heating water.

Advantages

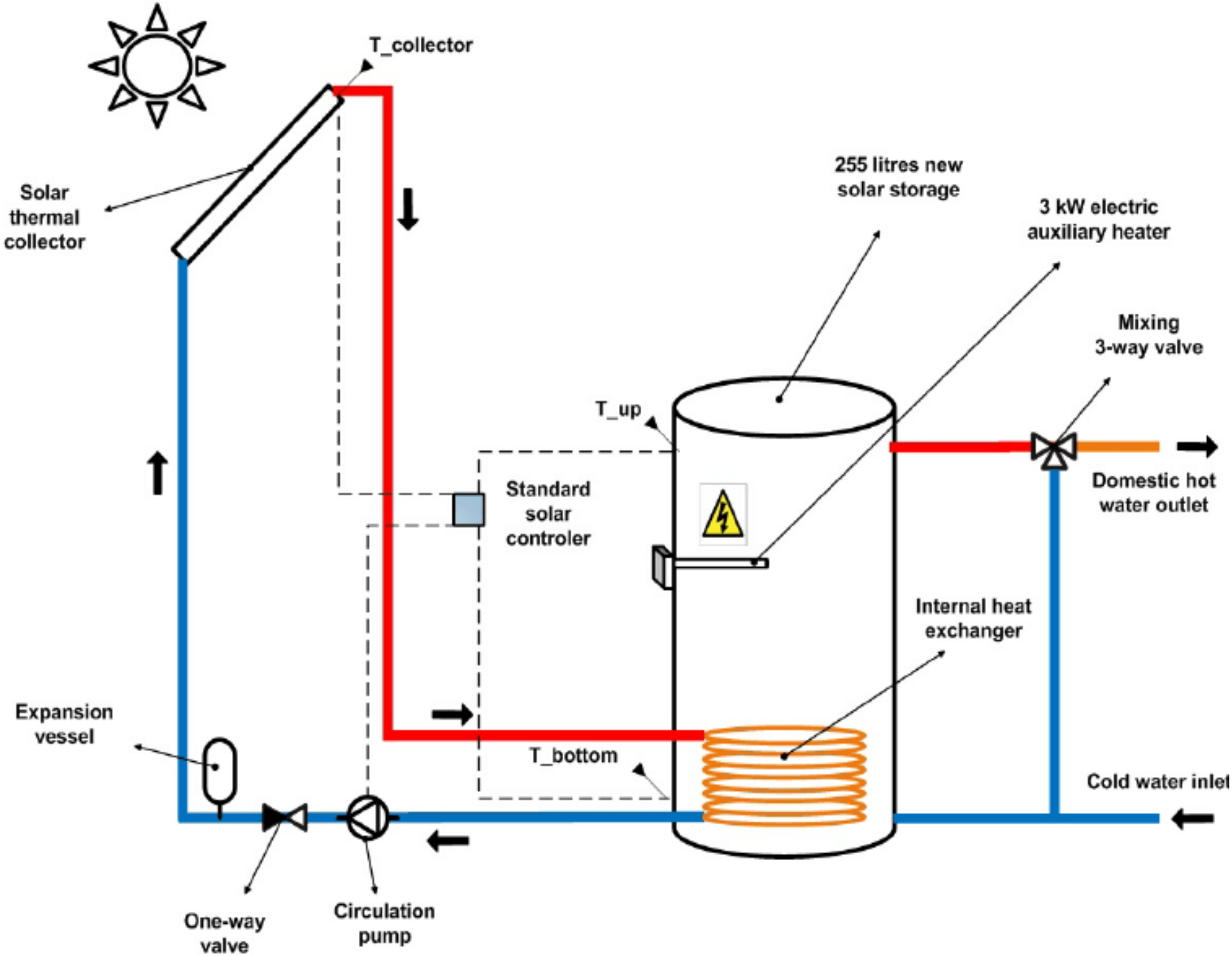
- **Environmental benefits**: Solar water heaters produce clean and renewable energy without emitting greenhouse gases or other pollutants.
- **Less space**: Solar thermal panels are usually less space-intensive than photovoltaic panels. Fewer are required to heat water than in an array used to produce sufficient power for a home.
- **Energy independence**: These systems supply up to **80% of the hot water needed** for a home, reducing dependence on utility companies.
- **Low maintenance**: After installation, little maintenance is required, and a solar water heater can run for up to **20 years**.
- **Financial incentives**: Solar water heaters may qualify for a tax credit via the **Residential Clean Energy Credit**.

Disadvantages

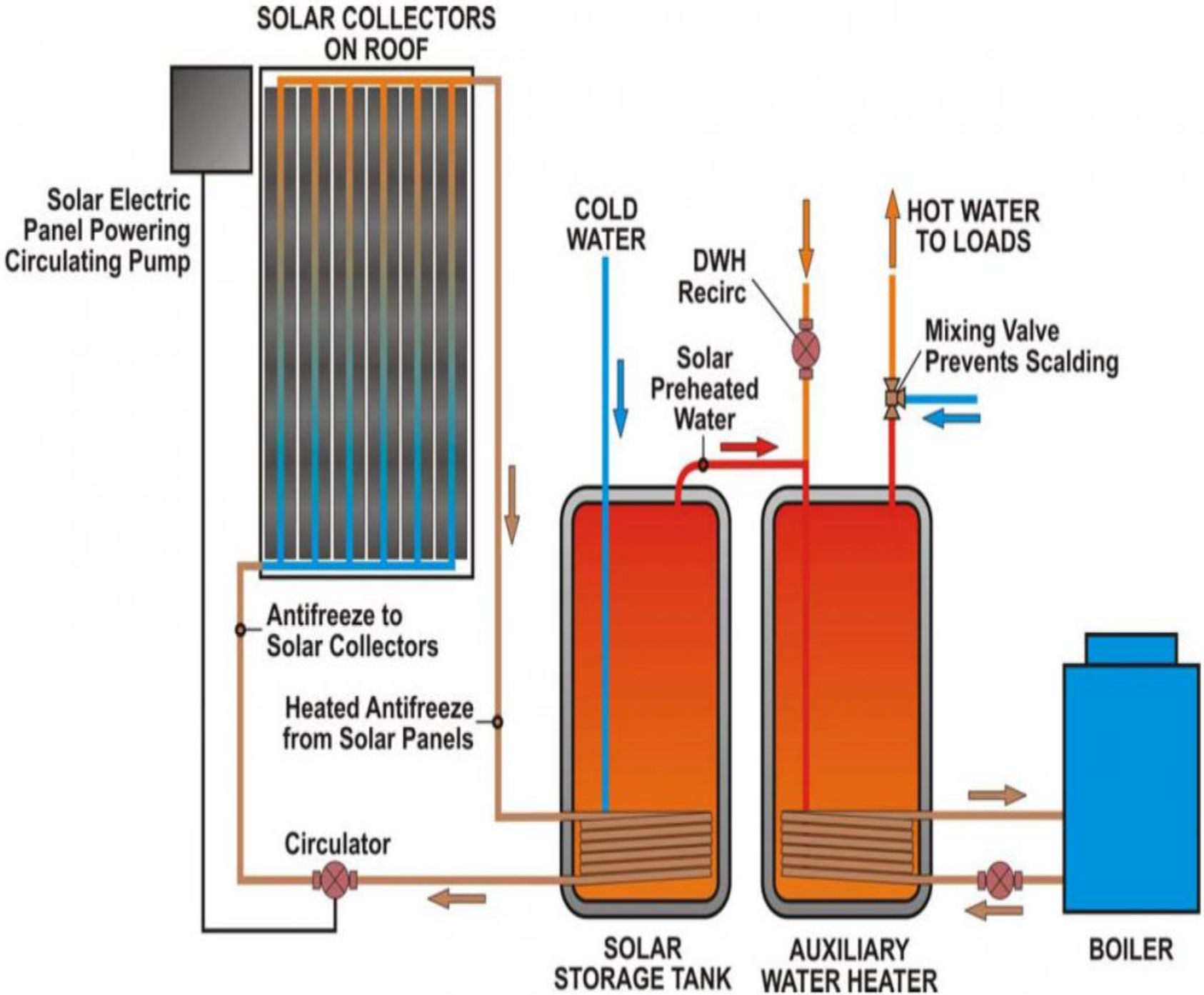
- Solar collectors require sufficient roof space.
- Solar water heaters require direct sunlight to function (i.e., they don’t function properly on cloudy, rainy, or foggy days).
- Annual maintenance is recommended to check the pump and other components.
- Installation requires the use of a new hot water cylinder.
- The **upfront cost** of purchasing and installing a solar water heater can be high compared to conventional water heating systems.

Active Solar Water Heating Systems

- **Direct circulation systems**
- Pumps circulate household water through the collectors and into the home.
- They work well in climates where it rarely freezes.
- Direct systems circulate water through solar collectors where it is heated by the sun.
- The heated water is then stored in a tank, sent to a tankless water heater, or used directly.
- These systems are preferable in climates where it rarely freezes.
- Freeze protection is necessary in cold climates.


T_collector
Solar thermal collector
255 litres new solar storage
3 kW electric auxiliary heater
Mixing 3-way valve
T_up
Standard solar controler
Domestic hot water outlet
Internal heat exchanger
Expansion vessel
T_bottom
Cold water inlet
One-way valve
Circulation pump

- **Indirect circulation systems**
  Pumps circulate a non-freezing, heat-transfer fluid through the collectors and a heat exchanger.
- This heats the water that then flows into the home.
- They are popular in climates prone to freezing temperatures.

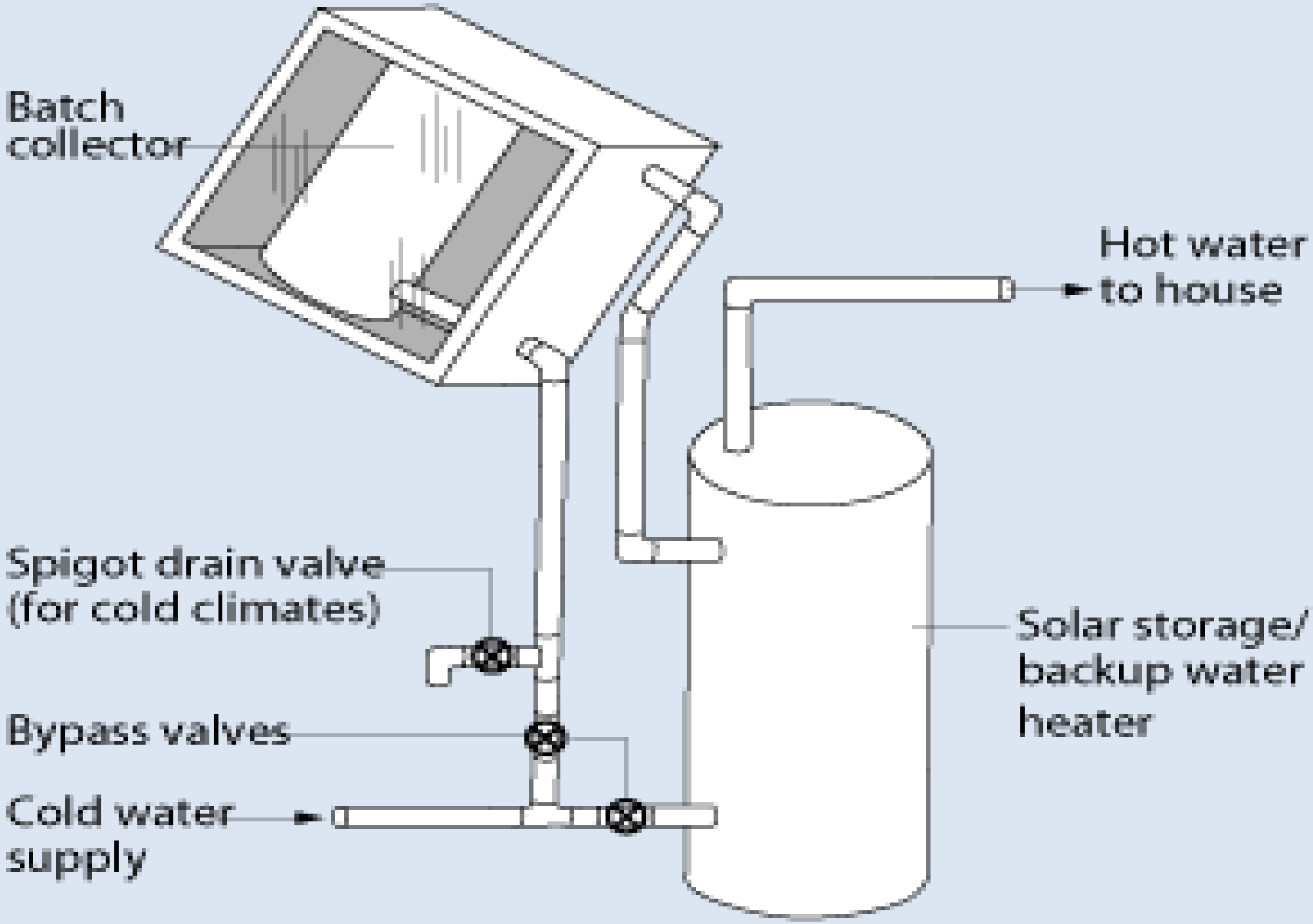

SOLAR COLLECTORS ON ROOF
Solar Electric Panel Powering Circulating Pump
COLD WATER
DWH Recirc
HOT WATER TO LOADS
Mixing Valve Prevents Scalding
Solar Preheated Water
Antifreeze to Solar Collectors
Heated Antifreeze from Solar Panels
Circulator
SOLAR STORAGE TANK
AUXILIARY WATER HEATER
BOILER

Passive Solar Water Heating Systems

- **Integral collector-storage passive systems**
  These consist of a storage tank covered with a transparent material to allow the sun to heat the water.
- Water from the tank then flows into the plumbing system.
- These work best in areas where temperatures rarely fall below freezing.
- They also work well in households with significant daytime and evening hot-water needs.

- **Thermosyphon systems**
  Water is heated in a collector on the roof and then flows through the plumbing system when a hot water faucet is opened.
- The majority of these systems have a 40 gallon capacity.

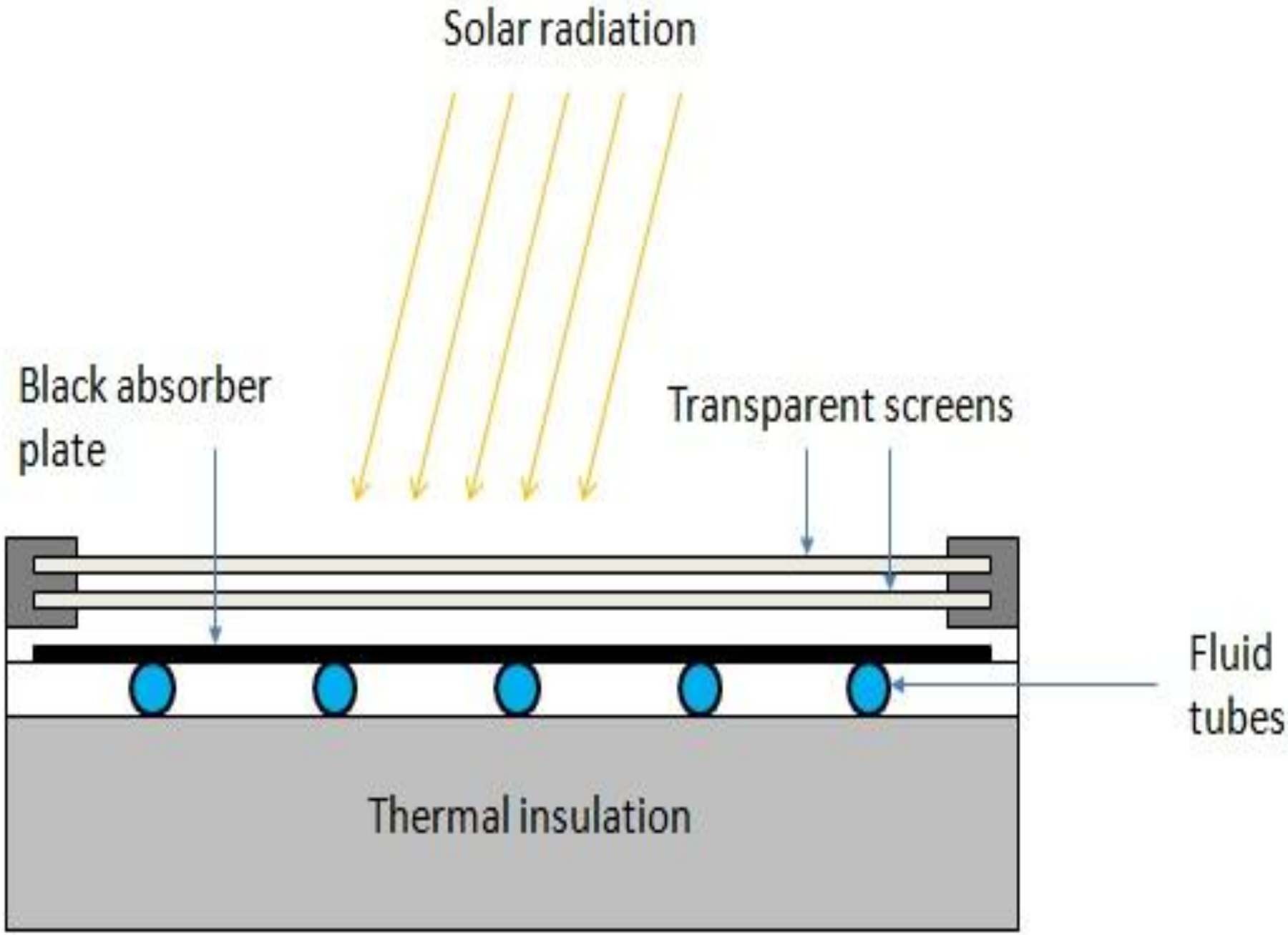
Passive, Batch Solar Water Heater



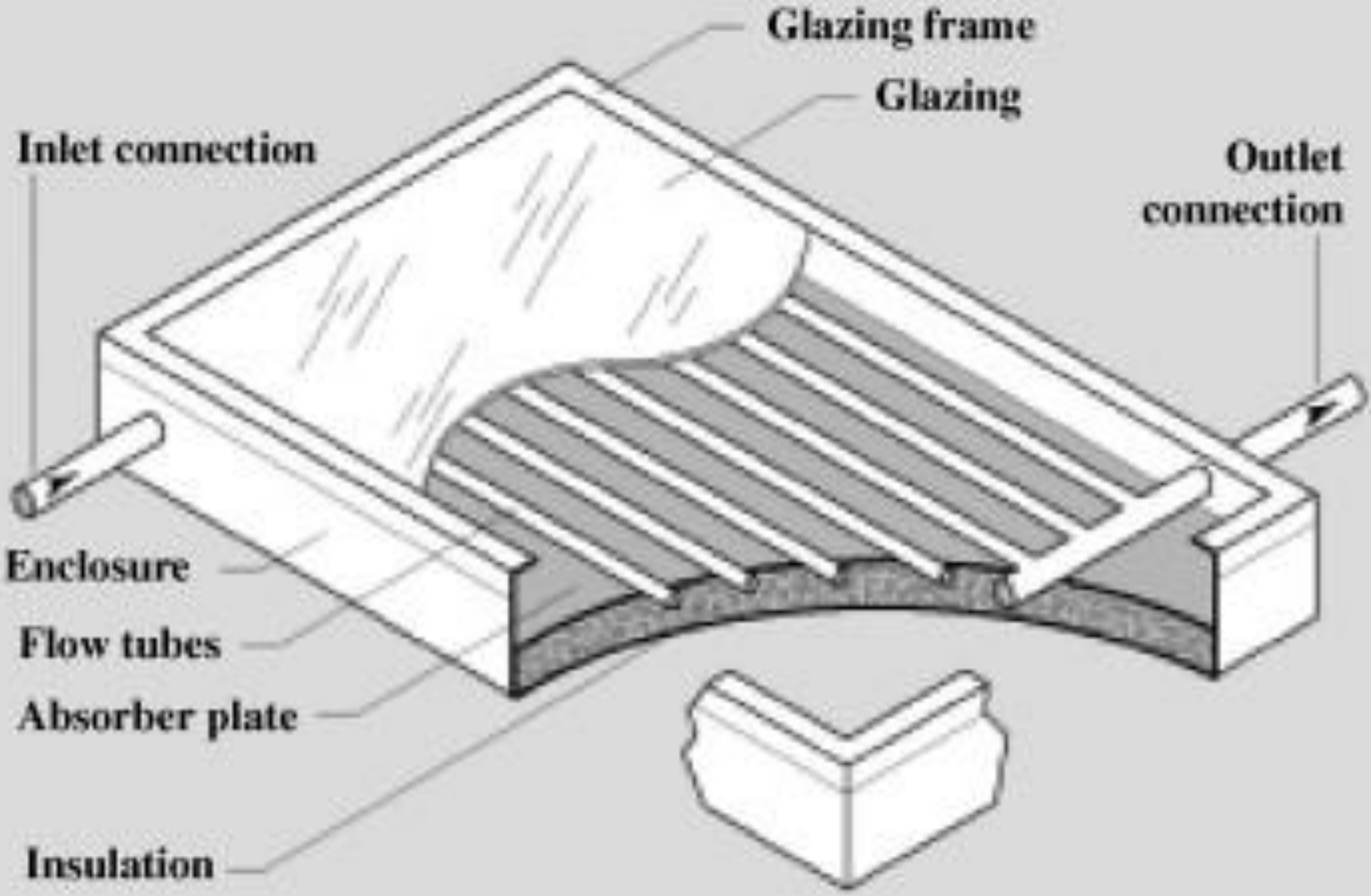
Flat-plate collector

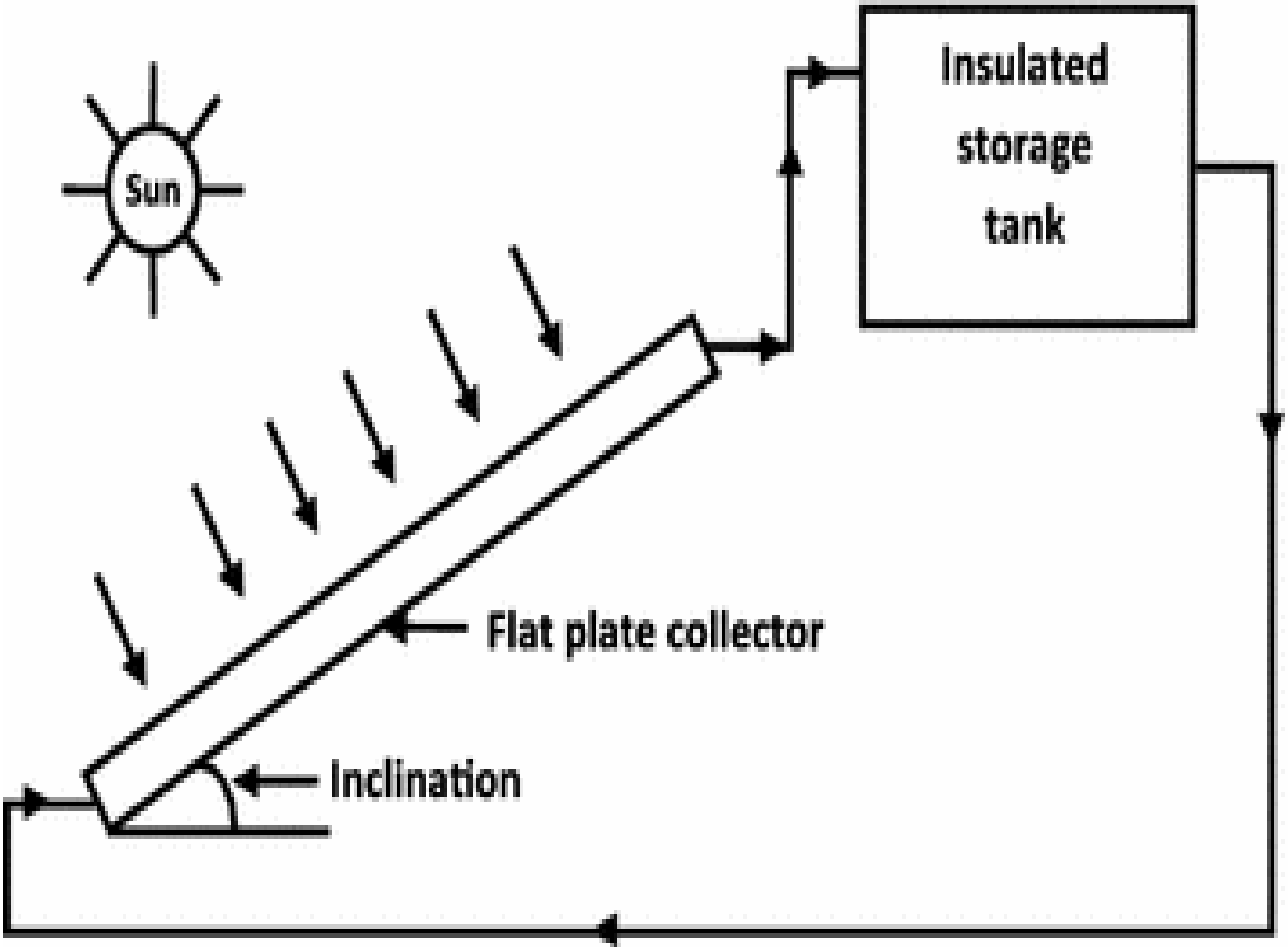
- These are the main components of a typical flat-plate solar collector:
- Black surface - absorbent of the incident solar energy
- Glazing cover - a transparent layer that transmits radiation to the absorber, but prevents radiative and convective heat loss from the surface
- Tubes containing heating fluid to transfer the heat from the collector
- Support structure to protect the components and hold them in place
- Insulation covering sides and bottom of the collector to reduce heat losses

- The flat-plate systems normally operate and reach the maximum efficiency within the temperature range from 30 to 80 ^{o}C ,however, some new types of collectors that employ vacuum insulation can achieve higher temperatures (up to 100 ^{o}C).
- Due to the introduction of selective coatings, the stagnant fluid temperature in flat-plate collectors has been shown to reach 200 ^{o}C.


Solar radiation
Black absorber plate
Transparent screens
Fluid tubes
Thermal insulation

Flat-Plate Collector

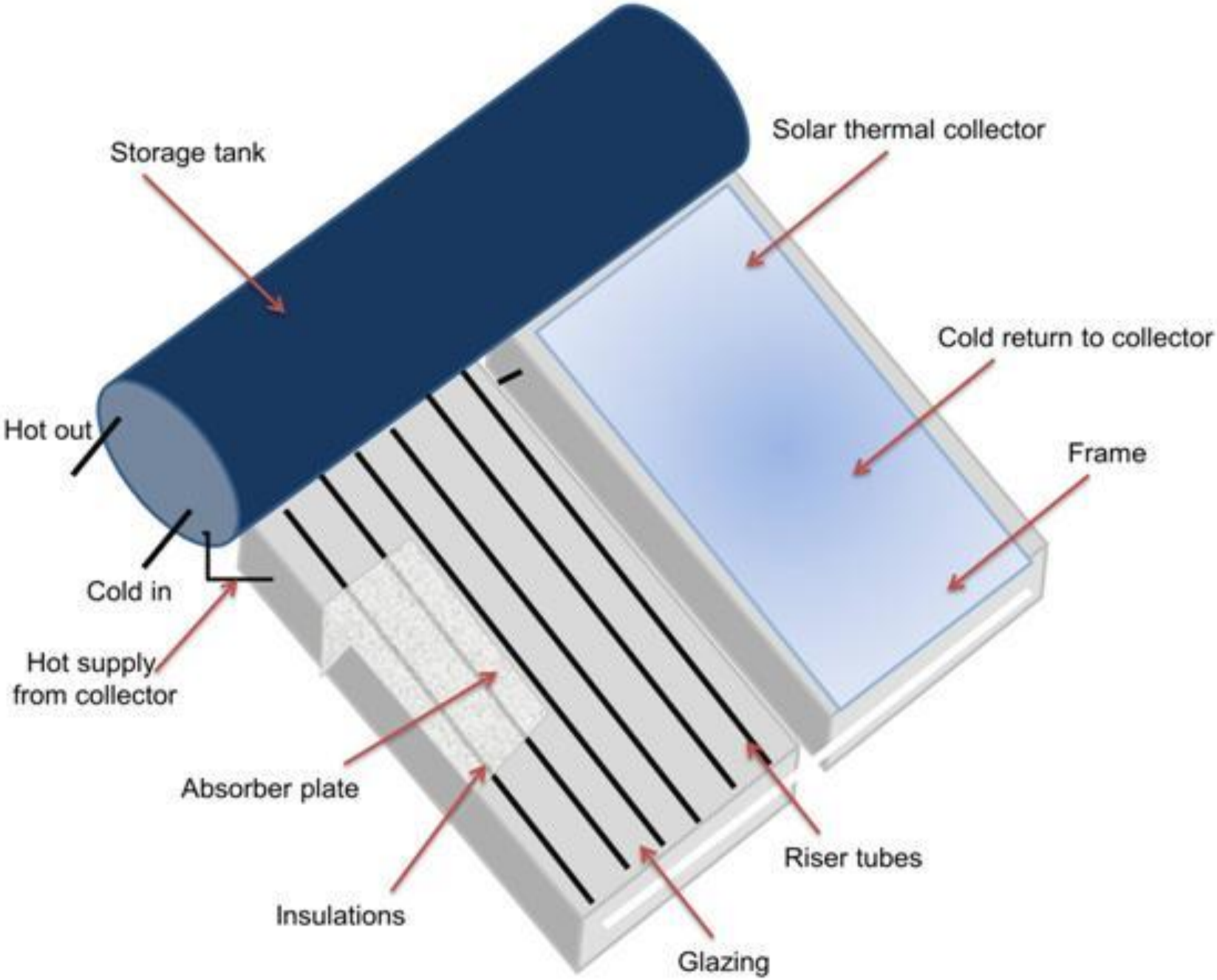



Sun
Insulated storage tank
Flat plate collector
Inclination

200LPD

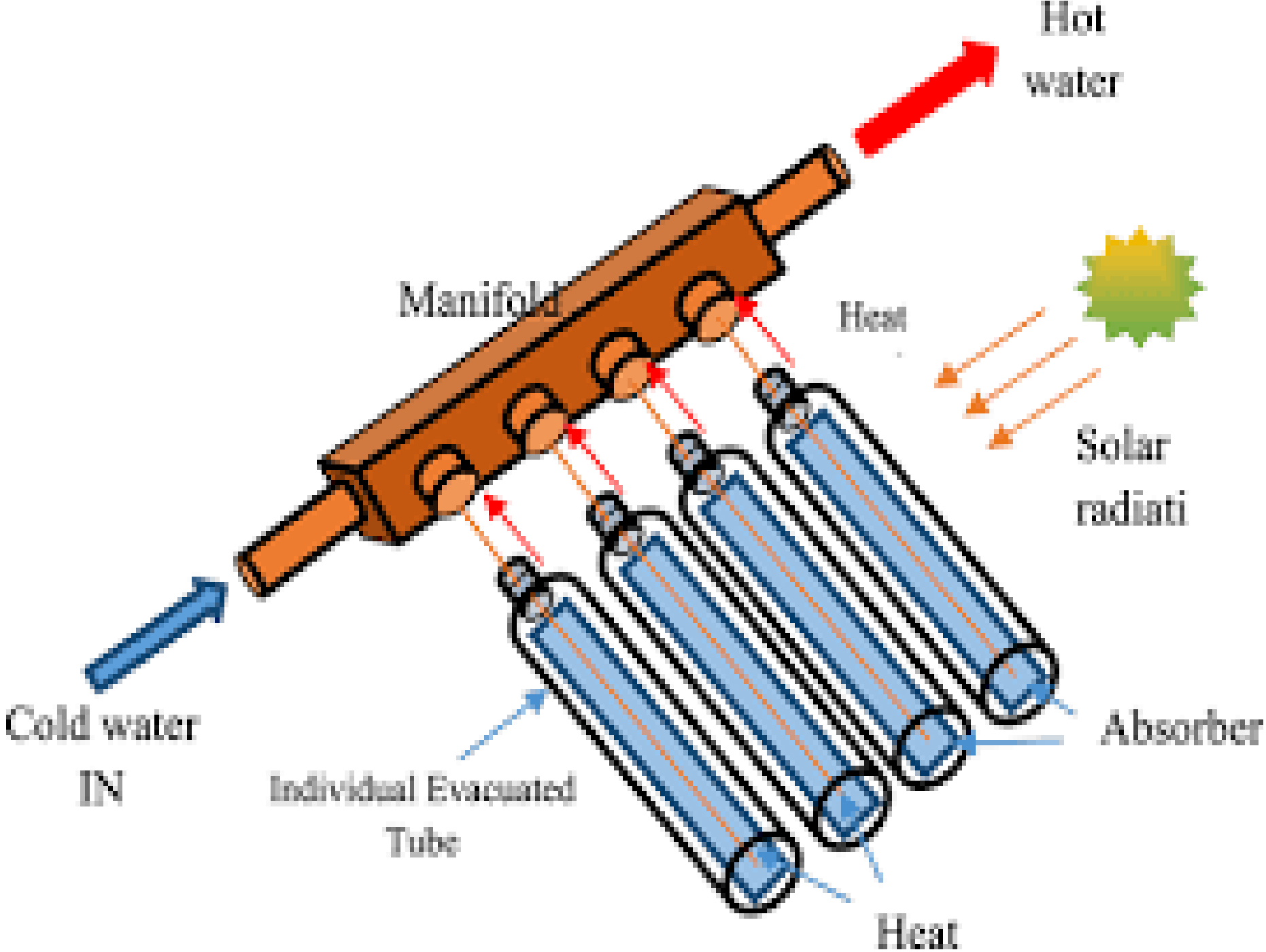
Integral collector-storage systems

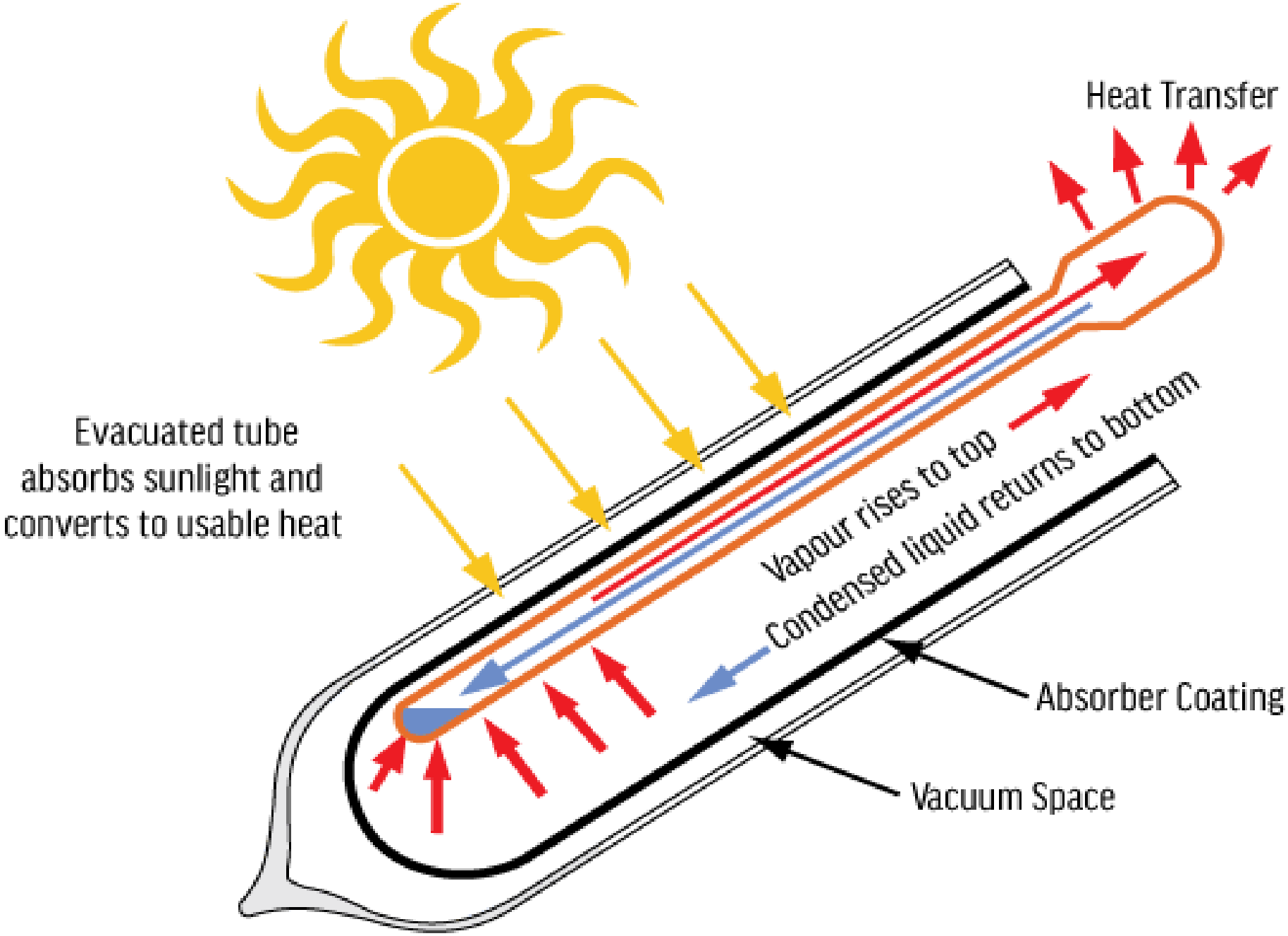
- Cold water first passes through the solar collector, which preheats the water.
- The water then continues on to the conventional backup water heater, providing a reliable source of hot water.
- They should be installed only in mild-freeze climates because the outdoor pipes could freeze in severe, cold weather.


Storage tank
Solar thermal collector
Cold return to collector
Hot out
Frame
Cold in
Hot supply
from collector
Absorber plate
Riser tubes
Insulations
Glazing

Evacuated-tube solar collectors

- They feature parallel rows of transparent glass tubes.
- Each tube contains a glass outer tube and metal absorber tube attached to a fin.
- The fin's coating absorbs solar energy but inhibits radiative heat loss.
- These collectors are used more frequently for U.S. commercial applications.


Hot
water
Manifold
Heat
Solar
radiati
Cold water
IN
Absorber
Individual Evacuated
Tube
Heat


Heat Transfer
Evacuated tube absorbs sunlight and converts to usable heat
Vapour rises to top
Condensed liquid returns to bottom
Absorber Coating
Vacuum Space

Solar Air Heater

- Solar air heating is a solar thermal technology in which the energy from the sun, insolation, is captured by an absorbing medium and used to heat air.
- Solar air heating is a renewable energy heating technology used to heat or condition air for buildings or process heat applications.

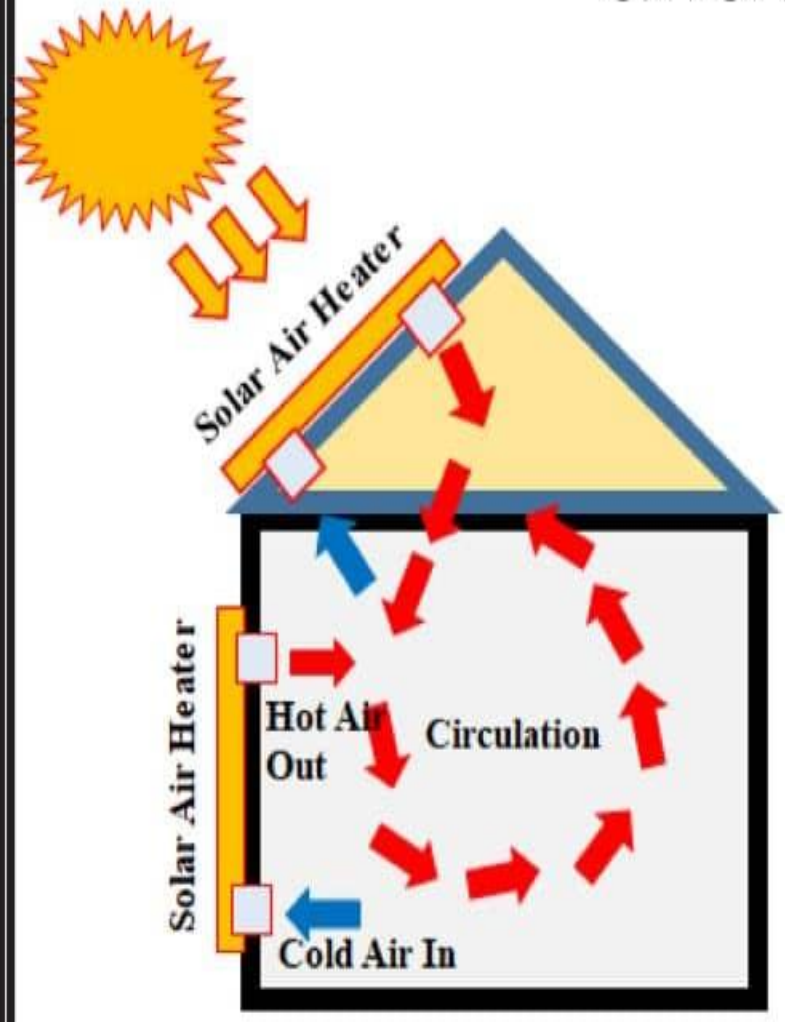
Advantages

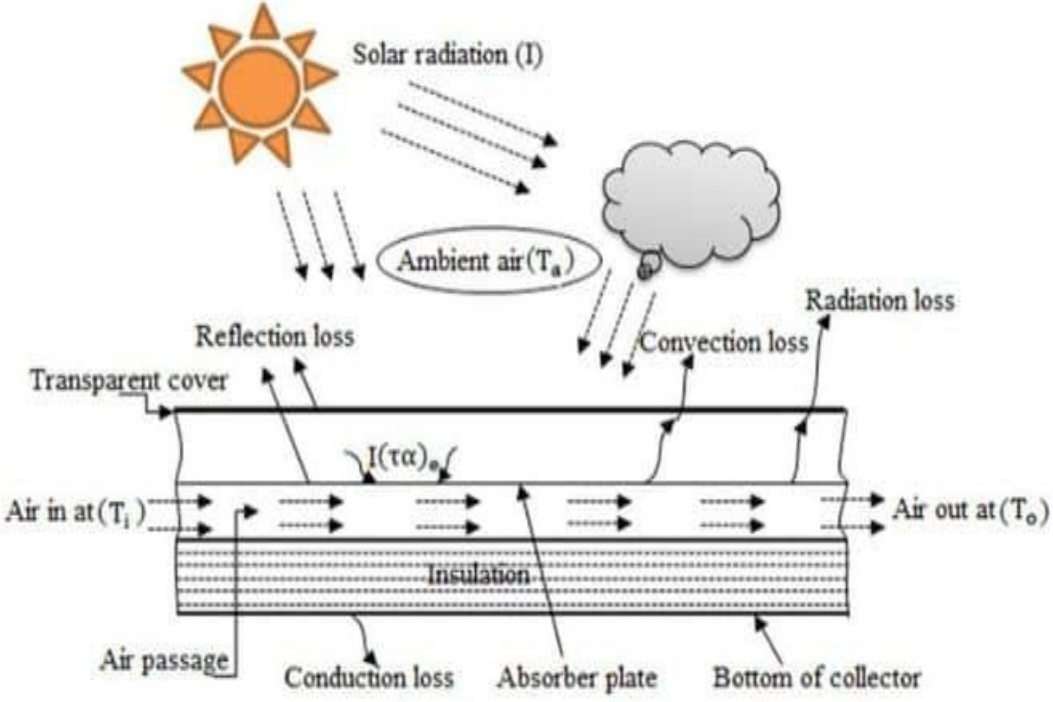
- Solar air heaters use air directly as the working substance, eliminating the need for complicated heat transfer systems.
- Unlike solar water heaters, solar air heaters do not face corrosion problems because they do not involve water.
- Air leakage from the ducts in solar air heaters does not cause significant issues, unlike water leakage in solar water heaters.
- Solar air heaters virtually do not experience freezing problems, as they do not use water-based working fluids like solar water heaters.
- The pressure inside the collector of a solar air heater remains low, allowing for a simpler and more cost-effective design with less material.
- Solar air heaters can be designed using cheaper and lesser amounts of material compared to solar water heaters, making them simpler to use and more economical.

Disadvantages

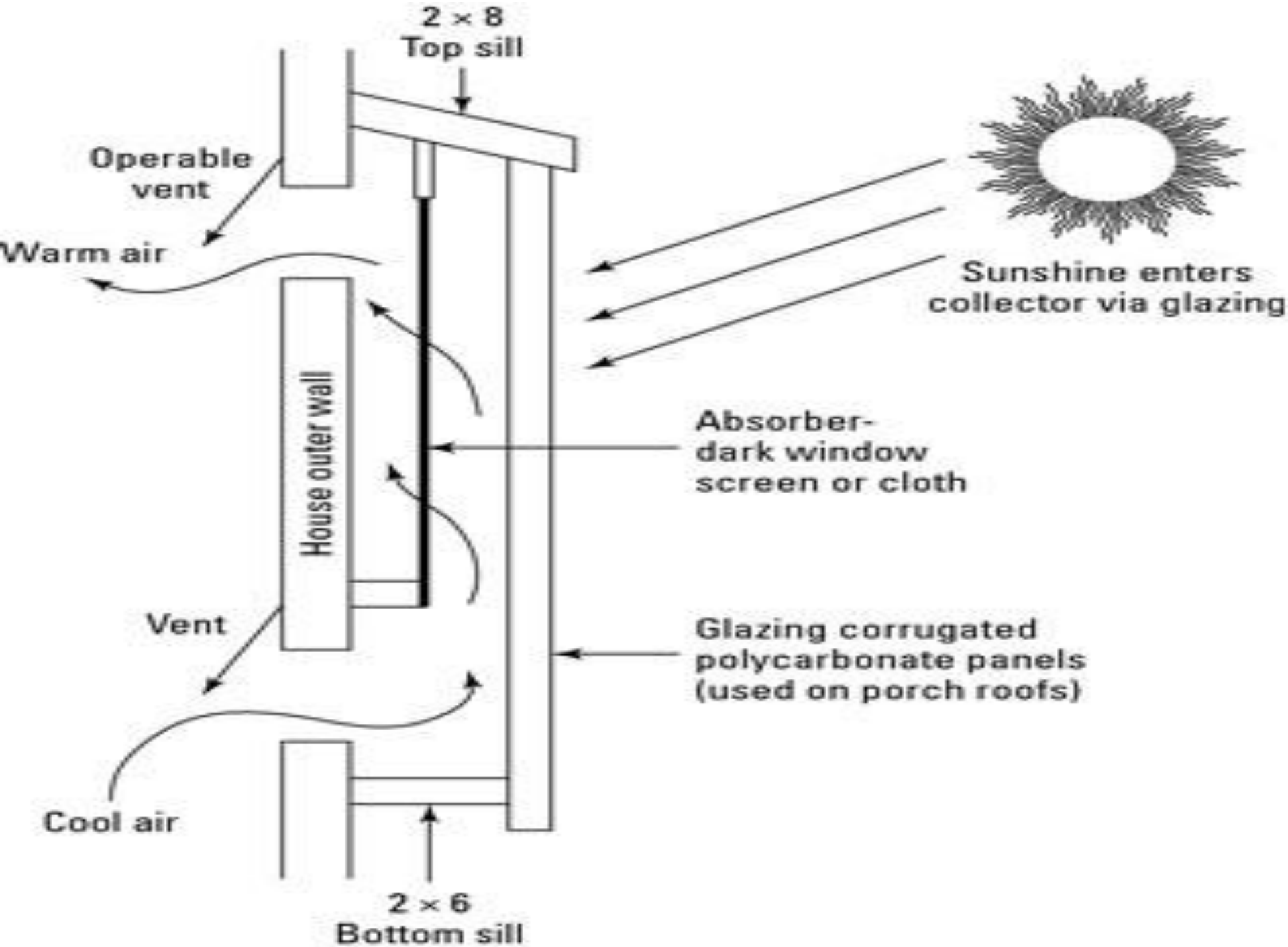
- Air has relatively poor heat transfer properties, so extra measures are needed to enhance its heat transfer efficiency.
- Air is not very dense, which means that a larger volume of air needs to be processed to achieve significant heating.
- Air cannot store heat well due to its low thermal capacity, making it unsuitable for use as a storage fluid.
- If not designed properly, the cost of solar air heaters can be quite high, making it essential to consider efficient and cost-effective designs.

Solar Air Heaters | Classification, Working, and Applications

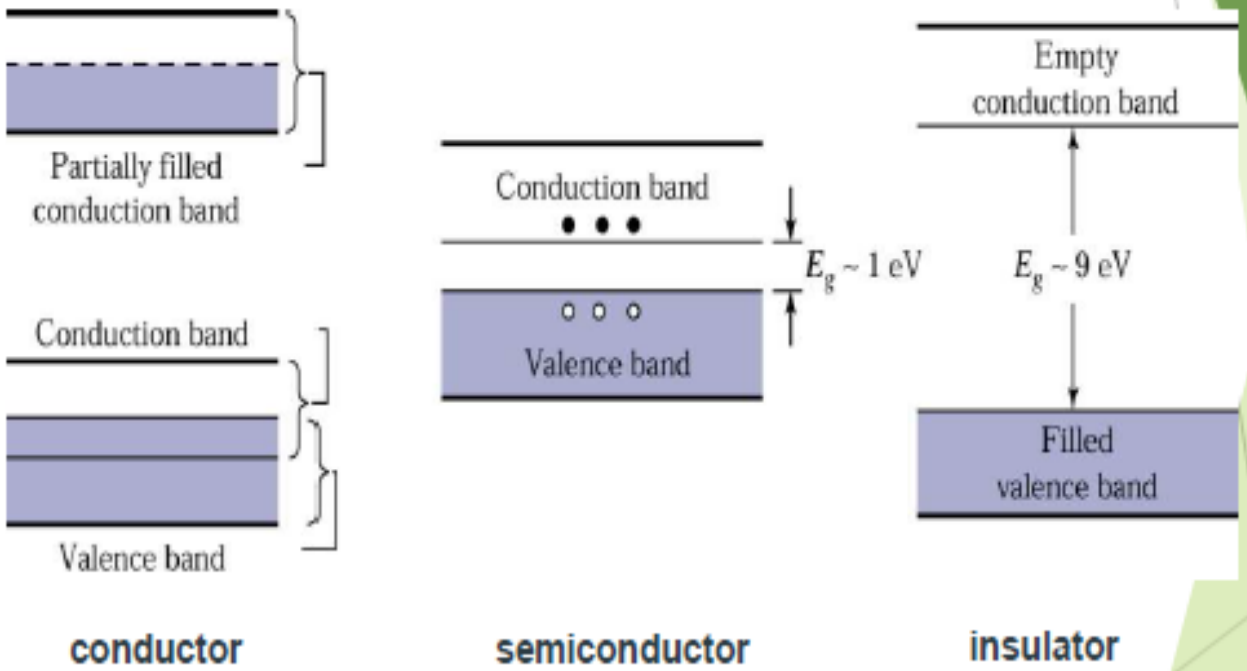




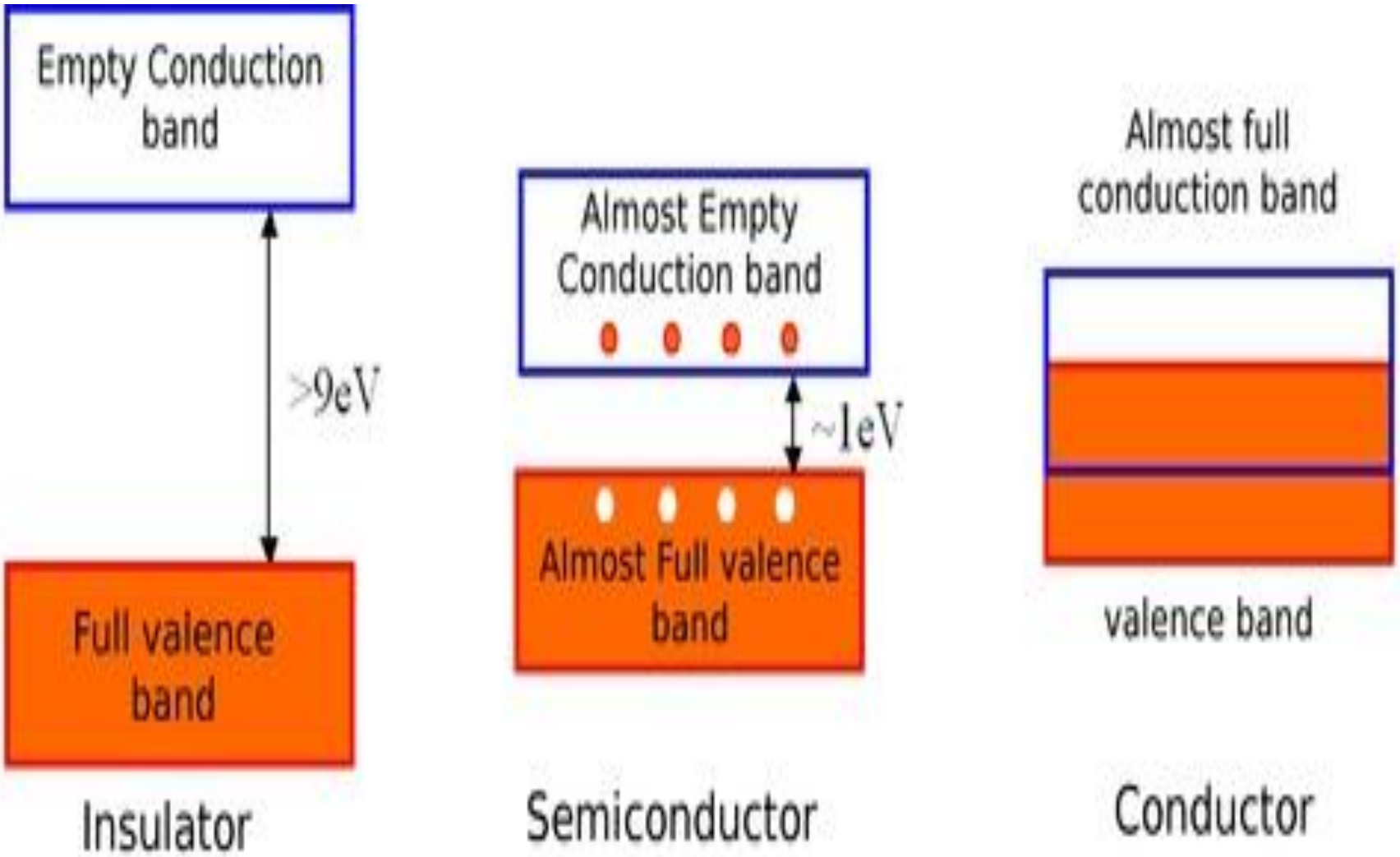
www.engineeringa2z.com


2 × 8
Top sill
Operable
vent
Warm air
Sunshine enters
collector via glazing
House outer wall
Absorber-
dark window
screen or cloth
Vent
Glazing corrugated
polycarbonate panels
(used on porch roofs)
Cool air
2 × 6
Bottom sill

Energy Band Diagram of a Conductor, Semiconductor and Insulator



Semiconductors are interested because their conductivity can be readily modulated (by impurity doping or electrical potential), offering a pathway to control electronic circuits.


Empty Conduction band
>9eV
Full valence band
Insulator
Almost Empty Conduction band
~1eV
Almost Full valence band
Semiconductor
Almost full conduction band
valence band
Conductor

Intrinsic Semiconductor

- Intrinsic semiconductor is extremely pure semiconductor.
- At room temperature, electron-hole pairs created in the intrinsic semiconductor crystal only due to thermal excitation.
- An intrinsic semiconductor does have a moderate conductivity of electricity due to the concentration of free electrons and holes thermally created in the crystal, at room temperature.

Extrinsic Semiconductor

- We can change the conductivity of the semiconductor significantly by adding some impurities in it.
- We refer to the process of adding impurities in a semiconductor for changing its conductive property, as doping.
- Added impurities in the semiconductor either increase the number of free electrons or increase the number of holes depending upon the type of impurities added.
- When we add penta-valent elements as impurities, the number of free electrons in the extrinsic semiconductor crystal increases.
- When we add trivalent elements as impurities the number of holes in the semiconductor increase.
- We call, the extrinsic semiconductor with more number of free electrons than holes as n-type semiconductor.
- We call the extrinsic semiconductor with more number of holes than the number of free electrons, as p-type semiconductor.
- So the amount of dopping determines the conductivity of the semiconductor.
- The type of dopped elements determines the type of semiconductor whether it is n-type or p-type in nature.

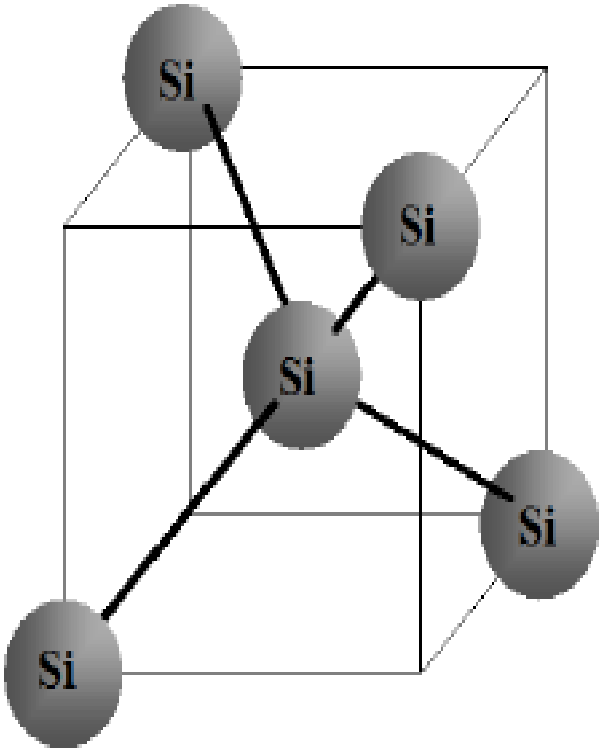
N Type

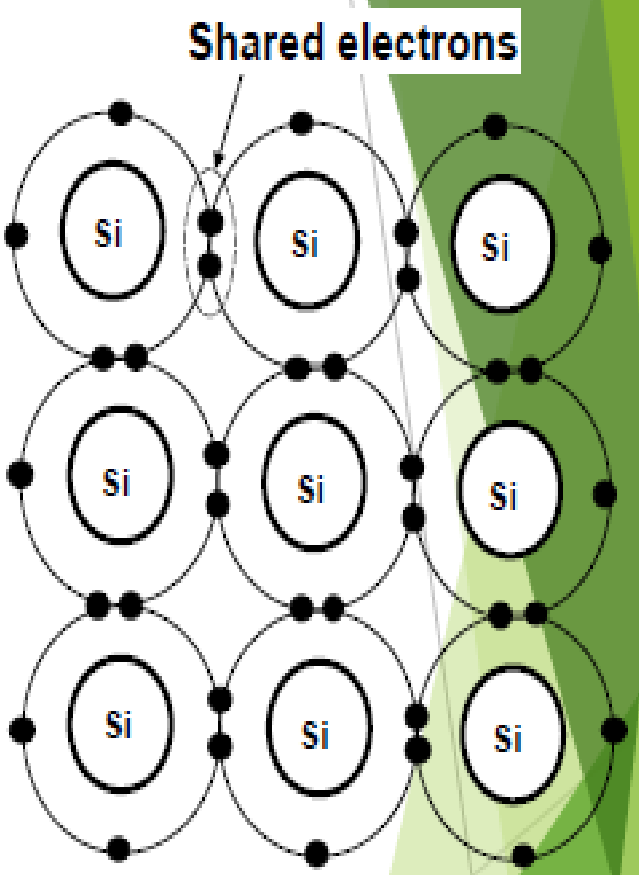
- **N-type (e.g. doped with Antimony)**
- These are materials which have **Penta-valent** impurity atoms (Donors) added and conduct by “electron” movement and are therefore called, **N-type Semiconductors**.
- In N-type semiconductors there are:
- 1. The Donors are positively charged.
- 2. There are a large number of free electrons.
- 3. A small number of holes in relation to the number of free electrons.
- 4. Doping gives:
  - positively charged donors.
  - negatively charged free electrons.
- 5. Supply of energy gives:
  - negatively charged free electrons.
  - positively charged holes.

P Type

- **P-type (e.g. doped with Boron)**
- These are materials which have **Trivalent** impurity atoms (Acceptors) added and conduct by “hole” movement and are therefore called, **P-type Semiconductors**.
- In these types of materials are:
- 1. The Acceptors are negatively charged.
- 2. There are a large number of holes.
- 3. A small number of free electrons in relation to the number of holes.
- 4. Doping gives:
  - negatively charged acceptors.
  - positively charged holes.
- 5. Supply of energy gives:
  - positively charged holes.
  - negatively charged free electrons.

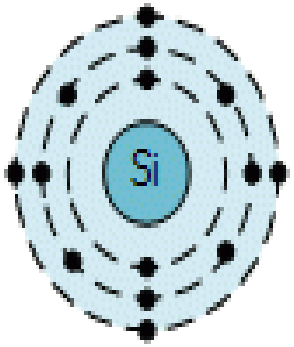
Silicon



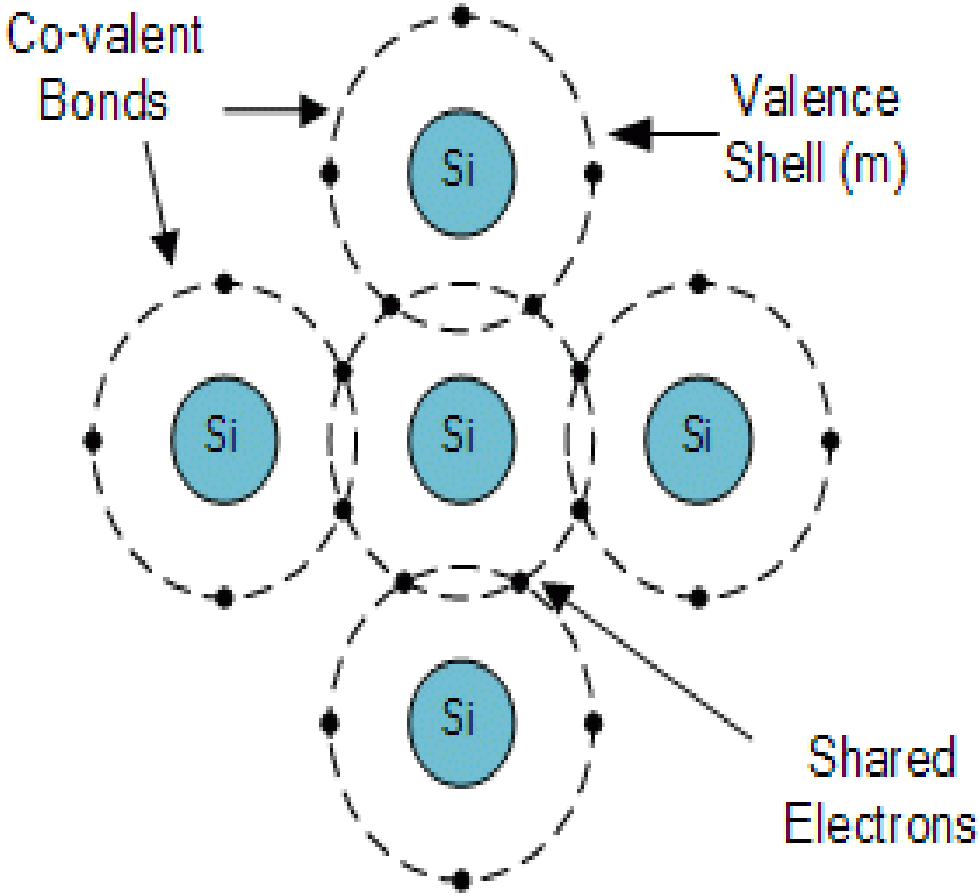


- Silicon is group IV element – with 4 electrons in their valence shell.
- When silicon atoms are brought together, each atom forms covalent bond with 4 silicon atoms in a tetrahedron geometry.

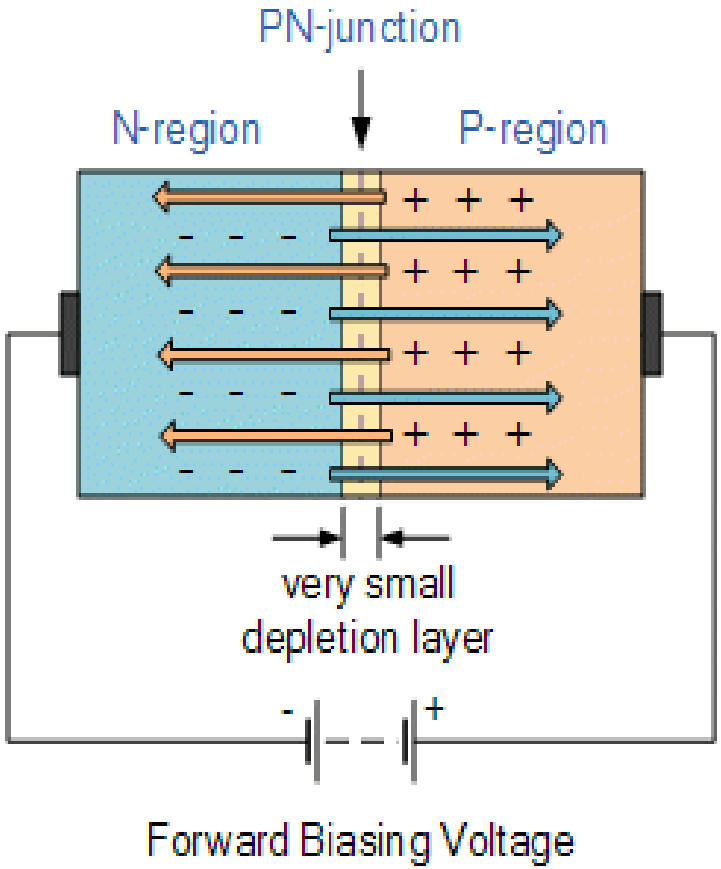
A Silicon Atom,
Atomic number = "14"

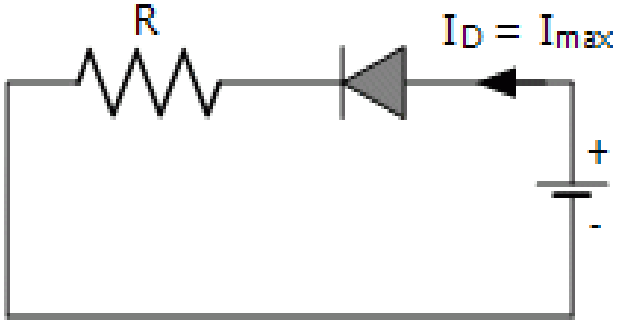


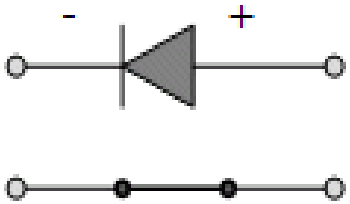
Silicon atom showing
4 electrons in its outer
valence shell (m)



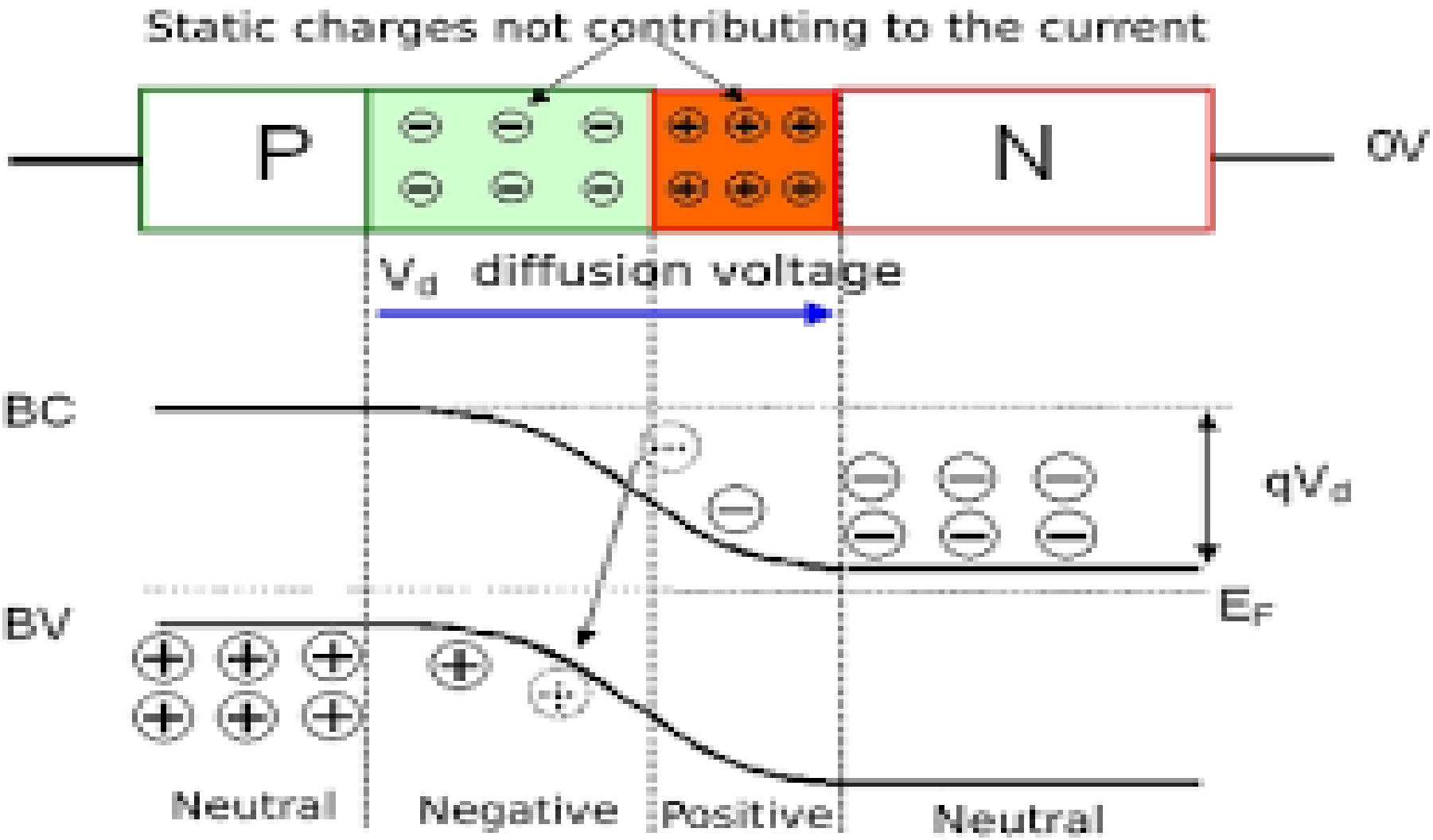
Silicon Crystal Lattice

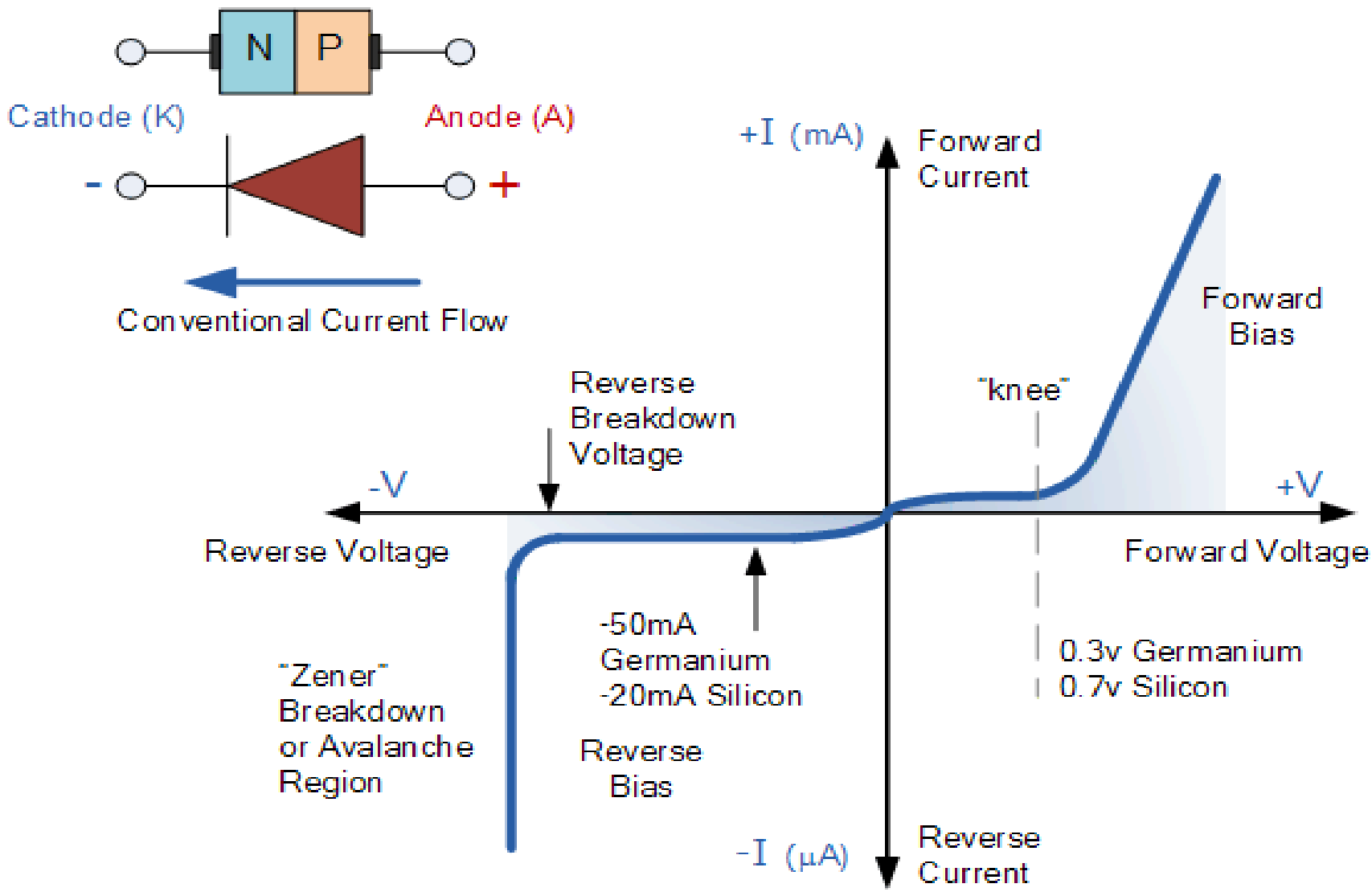

PN-junction
N-region
P-region
very small
depletion layer
-
+
Forward Biasing Voltage


R
$I_D = I_{max}$
+
-


-
+

Hole and Electron




N
P
Cathode (K)
Anode (A)
-
+
Conventional Current Flow
+I (mA)
Forward
Current
Forward
Bias
"knee"
Reverse
Breakdown
Voltage
-V
+V
Reverse Voltage
Forward Voltage
-50mA
Germanium
-20mA Silicon
0.3v Germanium
0.7v Silicon
"Zener"
Breakdown
or Avalanche
Region
Reverse
Bias
-I (μA)
Reverse
Current

- Silicon requires 1.1 eV for breaking any covalent bond in its crystal .
- Germanium requires 0.7 eV for the same purpose.
- $n = n_0 e^{-Eg/2.Kb.T}$
- Where,
- Eg = Energy bandgap
- K_b = Boltzmann's constants

$$I = I_0 \left(e^{\frac{qV}{kT}} - 1 \right)$$

where:

I = the net current flowing through the diode;

I_0 = "dark saturation current", the diode leakage current density in the absence of light;

V = applied voltage across the terminals of the diode;

q = absolute value of electron charge;

k = Boltzmann's constant; and

T = absolute temperature (K).

The Photovoltaic Effect

- The photovoltaic effect is the process of creating a voltage across charged materials that are exposed to electromagnetic radiation.
- Photons in sunlight impart their energy to excess charge carriers (electrons and holes) allowing them to freely move about the material.
- Charge opposition between the two materials creates an electrical field that provides momentum and direction to the free charge carriers, resulting in the flow of electrical current flow when the cell is connected to a load.

Activate
Go to PC s

Solar Photo voltaic

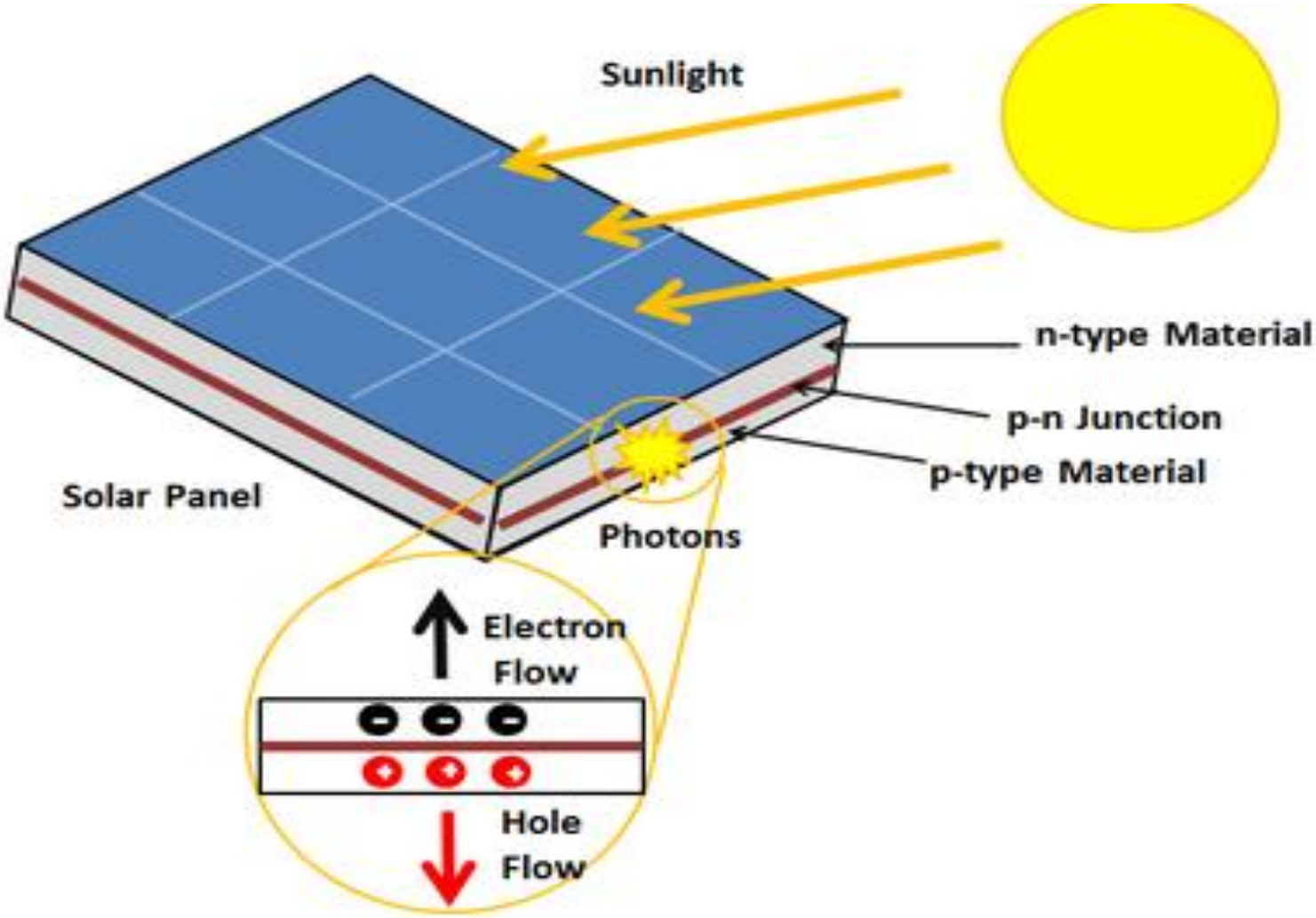
- A **solar cell**, or **photovoltaic cell**, is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect .

- It is a form of photoelectric cell, defined as a device whose electrical characteristics, such as current, voltage, or resistance, vary when exposed to light. Individual solar cell devices are often the electrical building blocks of photovoltaic modules, known colloquially as solar panels.

- The common single junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 to 0.6 volts.

- Solar cells are described as being photovoltaic, irrespective of whether the source is sunlight or an artificial light.

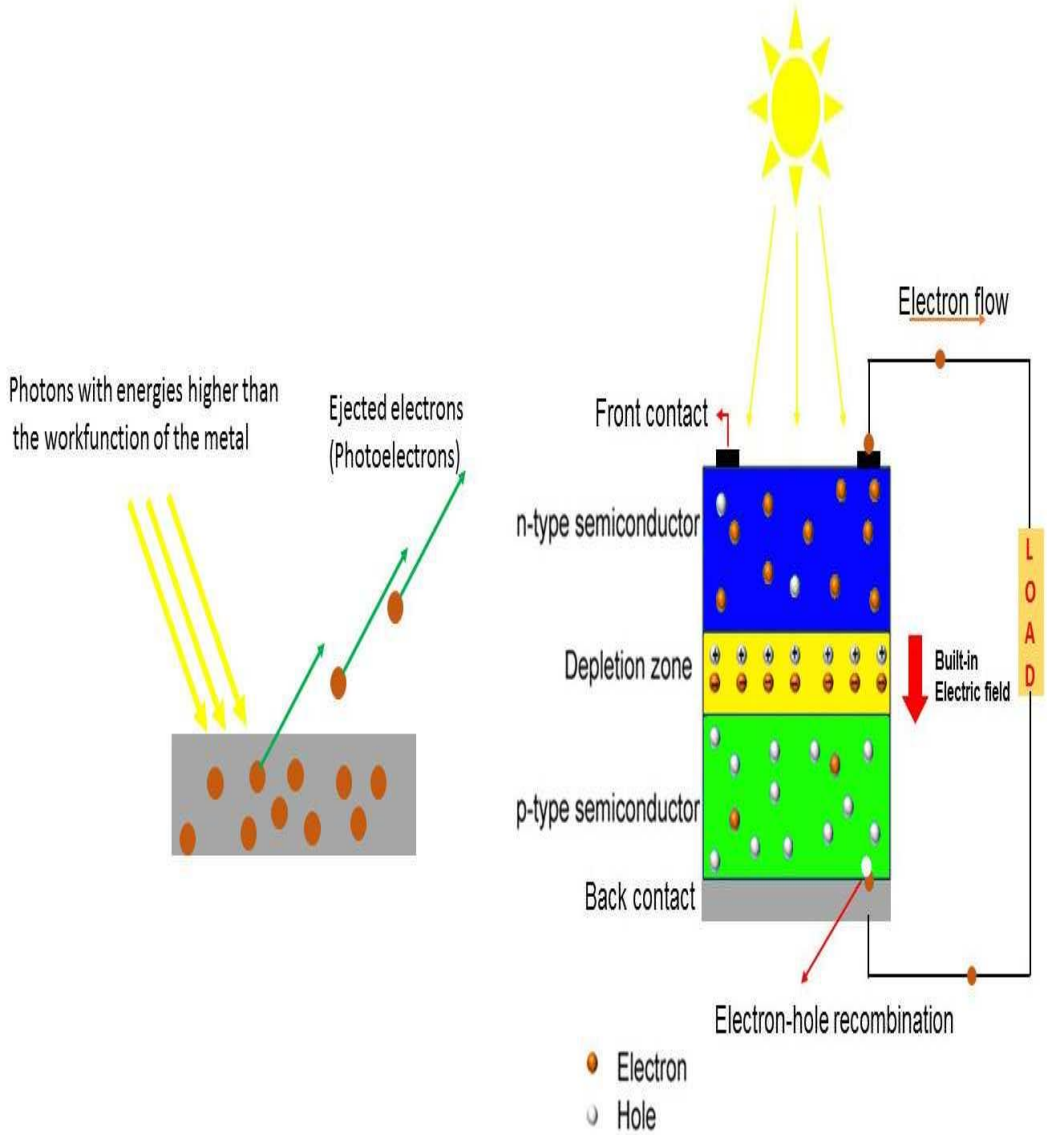
- In addition to producing energy, they can be used as a photo detector (for example infrared detectors), detecting light or other electromagnetic radiation near the visible range, or measuring light intensity.


Sunlight
n-type Material
p-n Junction
p-type Material
Solar Panel
Photons
Electron
Flow
Hole
Flow

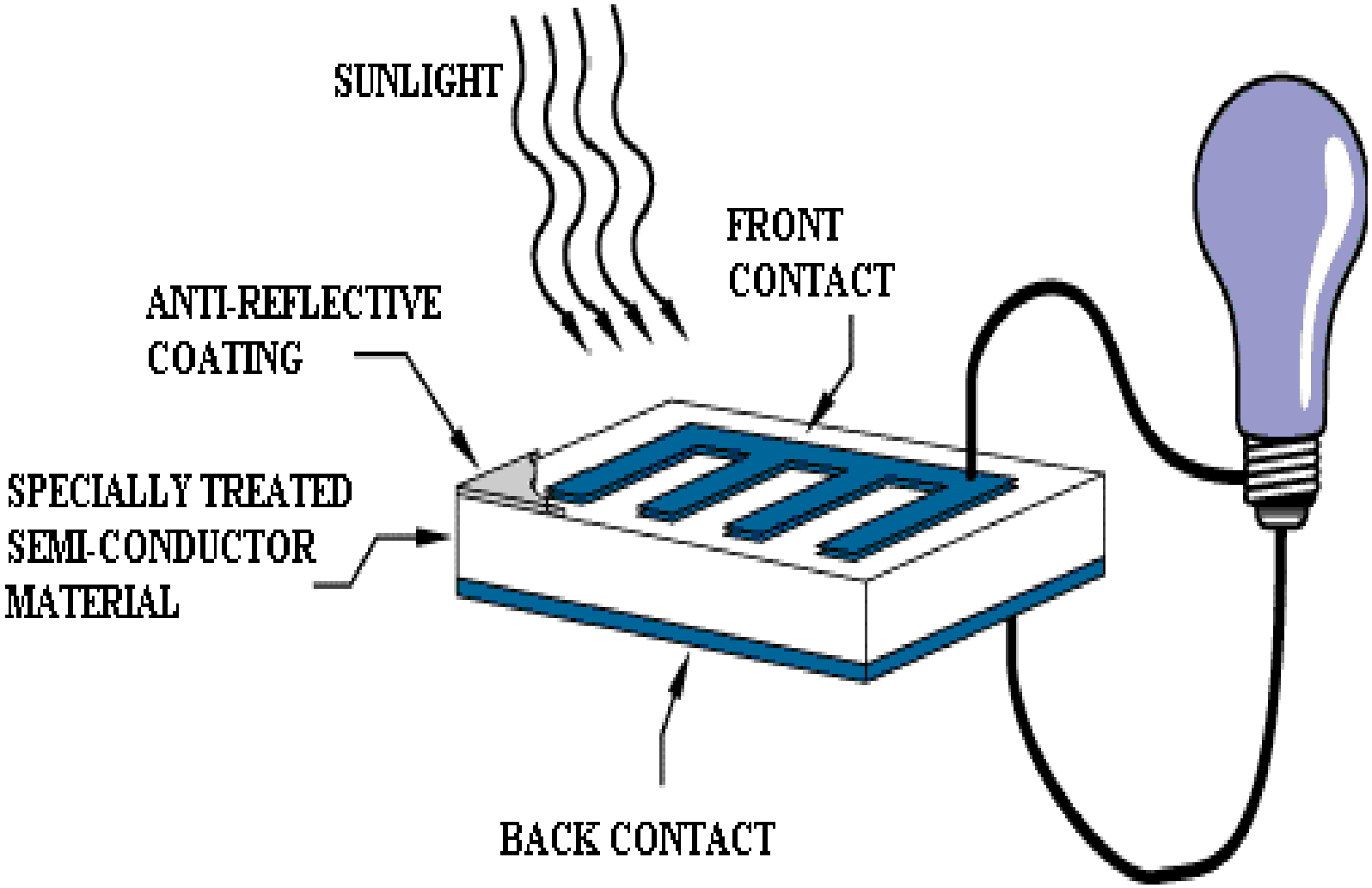
The solar cell works in several steps:

- Photons in sunlight hit the solar panel and are absorbed by semiconducting materials, such as doped silicon.
- Electrons are excited from their current molecular/atomic orbital.
- Once excited an electron can either dissipate the energy as heat and return to its orbital or travel through the cell until it reaches an electrode.
- Current flows through the material to cancel the potential and this electricity is captured.
- The chemical bonds of the material are vital for this process to work, and usually silicon is used in two layers, one layer being doped with boron, the other phosphorus.
- These layers have different chemical electric charges and subsequently both drive and direct the current of electrons.[1]
- An array of solar cells converts solar energy into a usable amount of direct current (DC) electricity.
- The most commonly known solar cell is configured as a large-area p–n junction made from silicon. Other possible solar cell types are organic solar cells, dye sensitized solar cells, perovskite solar cells, quantum dot solar cells etc.
- The illuminated side of a solar cell generally has a transparent conducting film for allowing light to enter into active material and to collect the generated charge carriers.
- Typically, films with high transmittance and high electrical conductance such as indium /tin oxide, conducting polymers or conducting nanowire networks are used for the purpose.

PHOTO VOLTAIC EFFECT

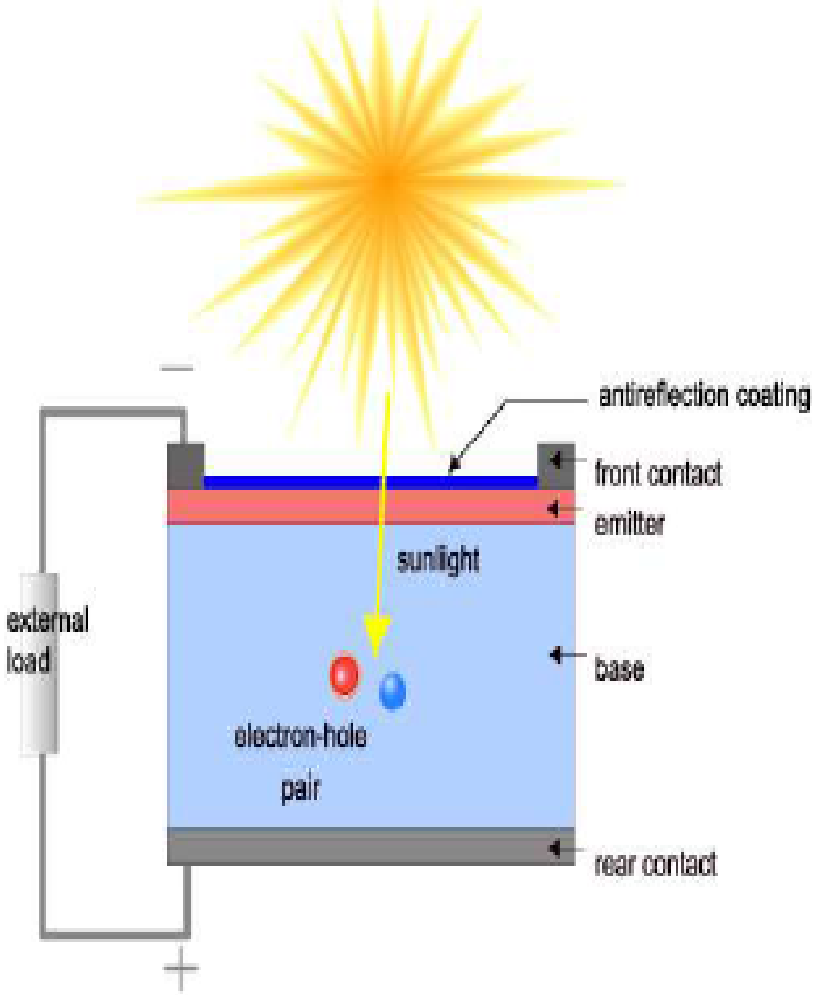


- The operation of a photovoltaic (PV) cell requires three basic attributes:
- The absorption of light, generating either electron-hole pairs .
- The separation of charge carriers of opposite types.
- The separate extraction of those carriers to an external circuit.

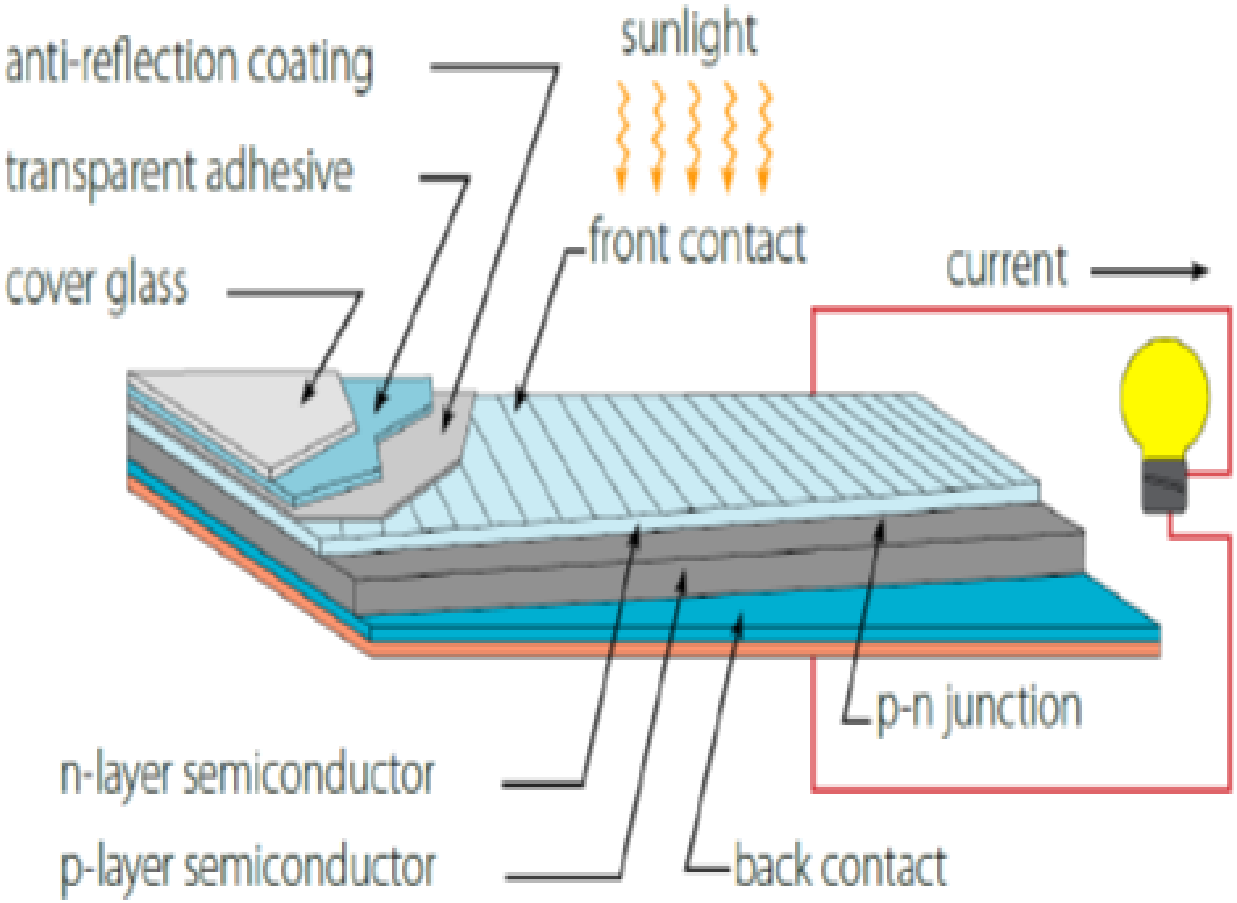

SUNLIGHT
FRONT CONTACT
ANTI-REFLECTIVE COATING
SPECIALLY TREATED SEMI-CONDUCTOR MATERIAL
BACK CONTACT

Photovoltaic Cell

- A PV cell is a light illuminated pn-junction diode which directly converts solar energy into electricity via the photovoltaic effect.
- A typical silicon PV cell is composed of a thin wafer consisting of an ultra-thin layer of phosphorus-doped (n-type) silicon on top of a thicker layer of boron-doped (p-type) silicon.
- When sunlight strikes the surface of a PV cell, photons with energy above the semiconductor bandgap impart enough energy to create electron-hole pairs.

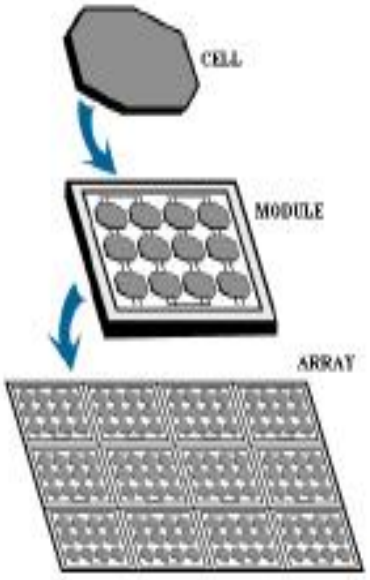


Structure of a Solar Cell



Hierarchy of PV

	Volt	Ampere	Watt	Size
Cell	0.5V	5-6A	2-3W	about 10cm
Module	20-30V	5-6A	100-200W	about 1m
Array	200-300V	50A-200A	10-50kW	about 30m



Array

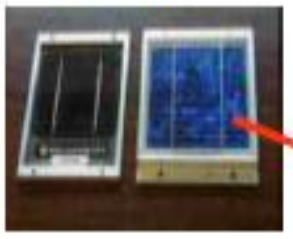
10 - 50 kW

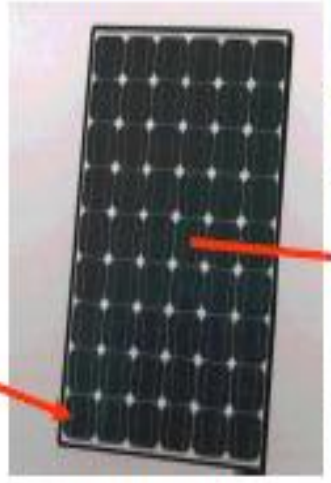
Module,Panel

100 - 200 W

Cell

2 – 3 W

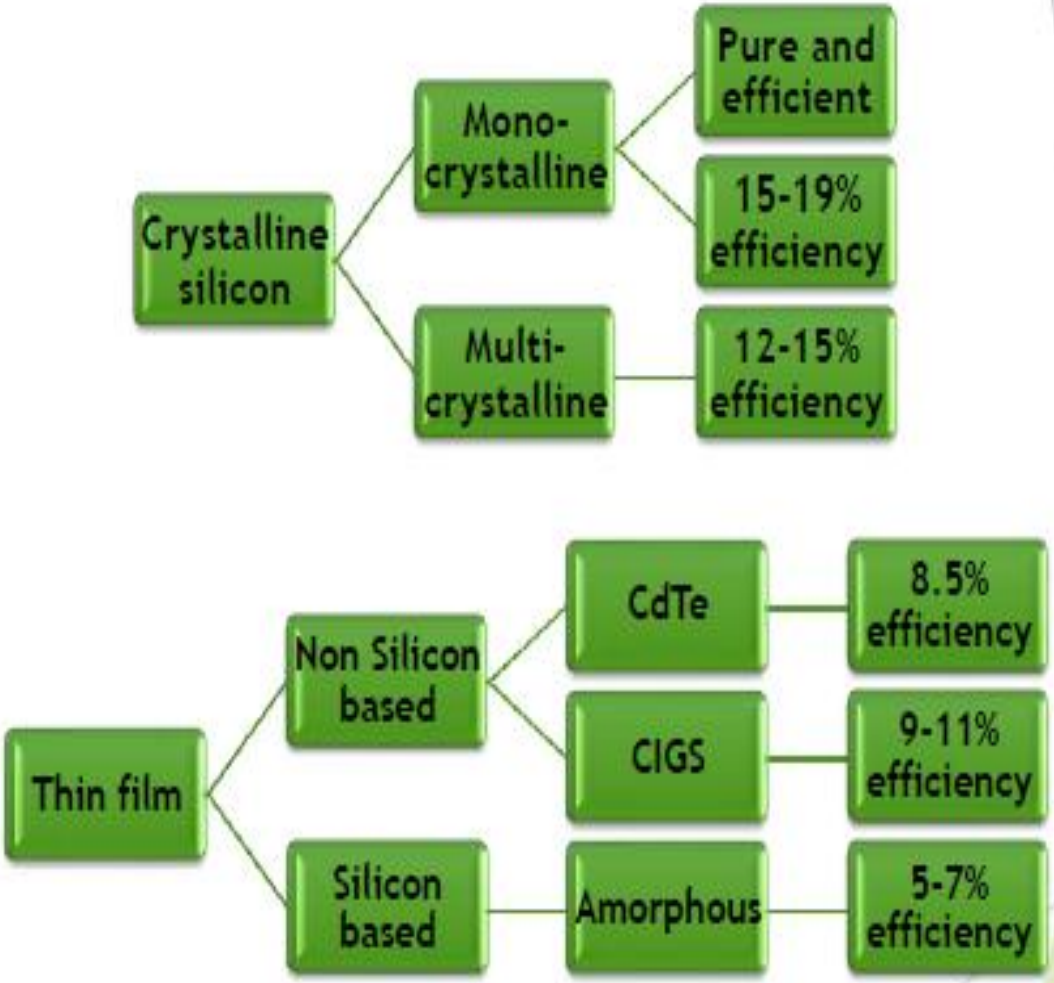




6x9=54 (cells)

100-300 (modules)

Cell Technologies



Three generations of solar cells

1. First Generation

- First generation cells consist of high quality and single junction devices.
- First Generation technologies involve high energy and labour inputs which prevent any significant progress in reducing production costs.

2. Second Generation

- Second generation materials have been developed to address energy requirements and production costs of solar cells.
- Alternative manufacturing techniques such as vapour deposition and electroplating are advantageous as they reduce high temperature processing significantly.
- Produced from cheaper polycrystalline materials and glass
- High optical absorption coefficients
- Bandgap suited to solar spectrum

SUNRATOR

3. Third Generation

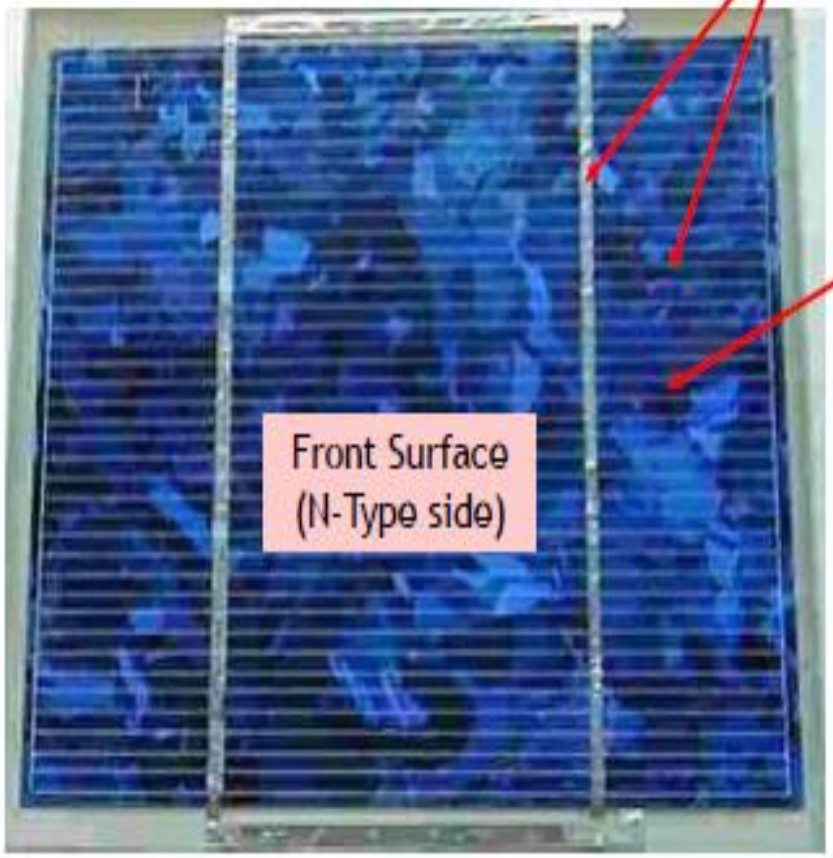
- Third generation technologies aim to enhance poor electrical performance of second generation (thin-film technologies) while maintaining very low production costs.
- Current research is targeting conversion efficiencies of 30-60% while retaining low cost materials and manufacturing techniques.
- They can exceed the theoretical solar conversion efficiency limit for a single energy threshold material, 31% under 1 sun illumination and 40.8% under the maximal artificial concentration of sunlight (46,200 suns).
- Approaches to achieving these high efficiencies including the use of multijunction photovoltaic cells, concentration of the incident spectrum, the use of thermal generation by UV light to enhance voltage or carrier collection, or the use of the infrared spectrum for night-time operation.

Monocrystalline Silicon Modules

- Most efficient commercially available module (14% - 17%)
- Most expensive to produce
- Circular (square-round) cell creates wasted space on module

Polycrystalline Silicon Modules

- Aluminum Electrode (Silver colored wire)
- To avoid shading, electrode is very fine.



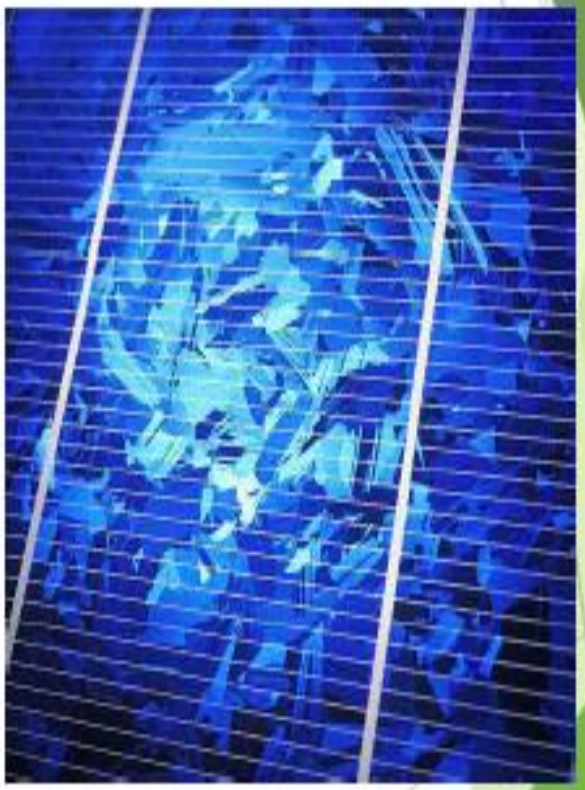
Anti reflection film (Blue colored film)

- Back surface is P-type.
- All back surface is aluminum electrode with full reflection.

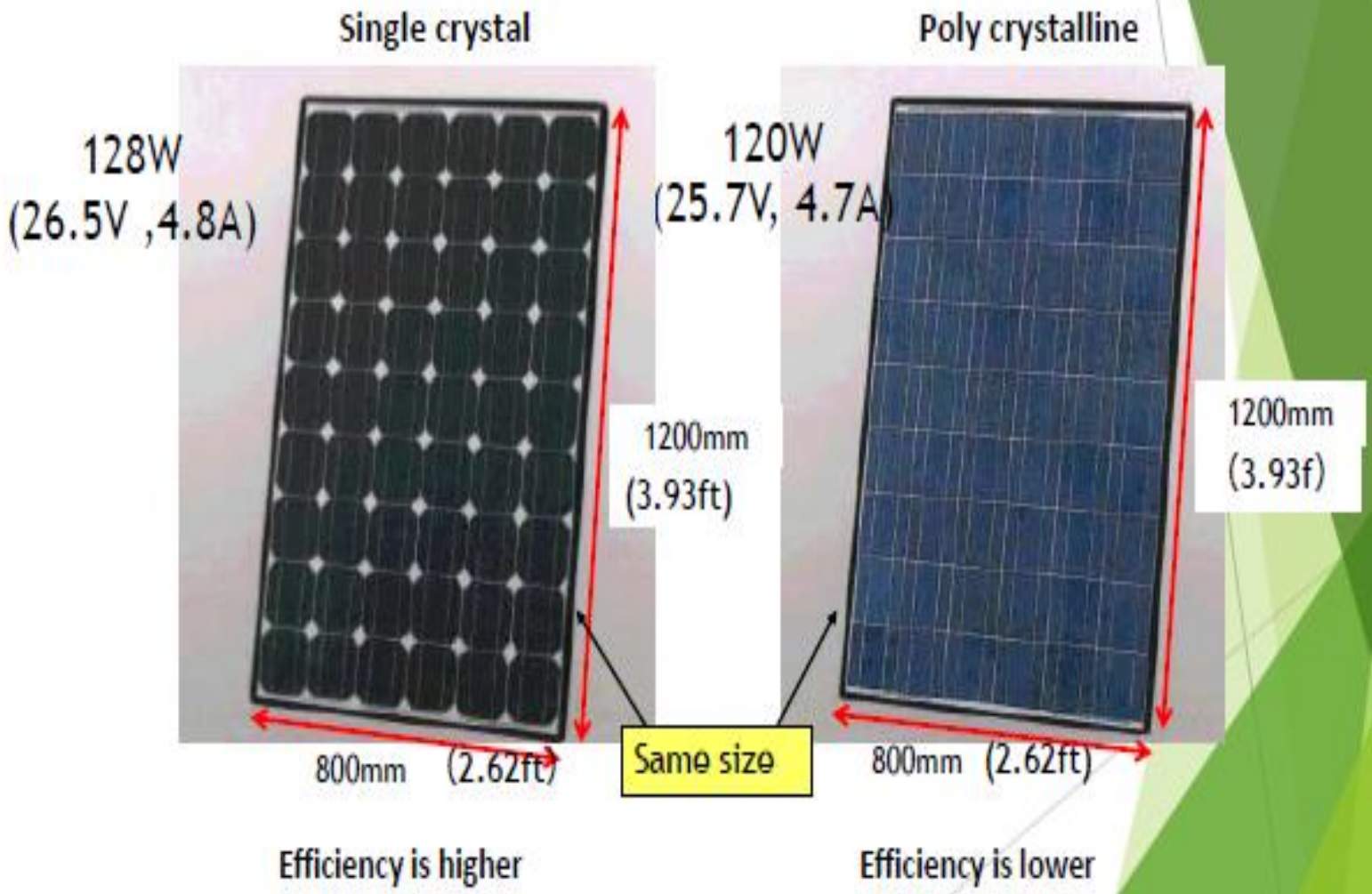
Poly Crystalline PV

Polycrystalline Silicon Modules

- Less expensive to make than single crystalline modules
- Cells slightly less efficient than a single crystalline (10% - 12%)
- Square shape cells fit into module efficiently using the entire space



PV Module (Single crystal, Poly crystalline Silicon)

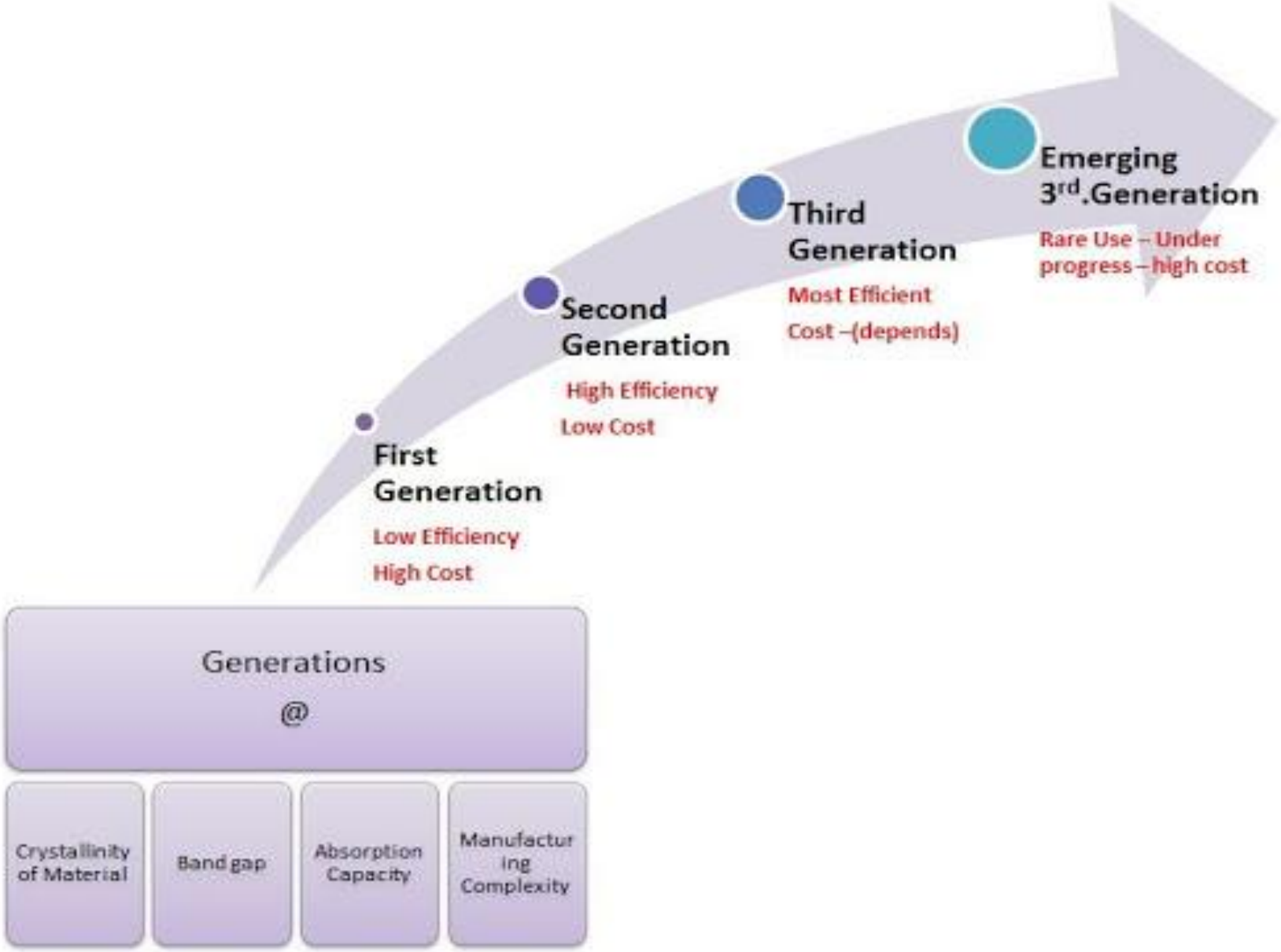


Amorphous Thin Film

- Most inexpensive technology to produce
- Metal grid replaced with transparent oxides
- Efficiency = 6 – 8 %
- Can be deposited on flexible substrates
- Less susceptible to shading problems
- Better performance in low light conditions that with crystalline modules

Comparision

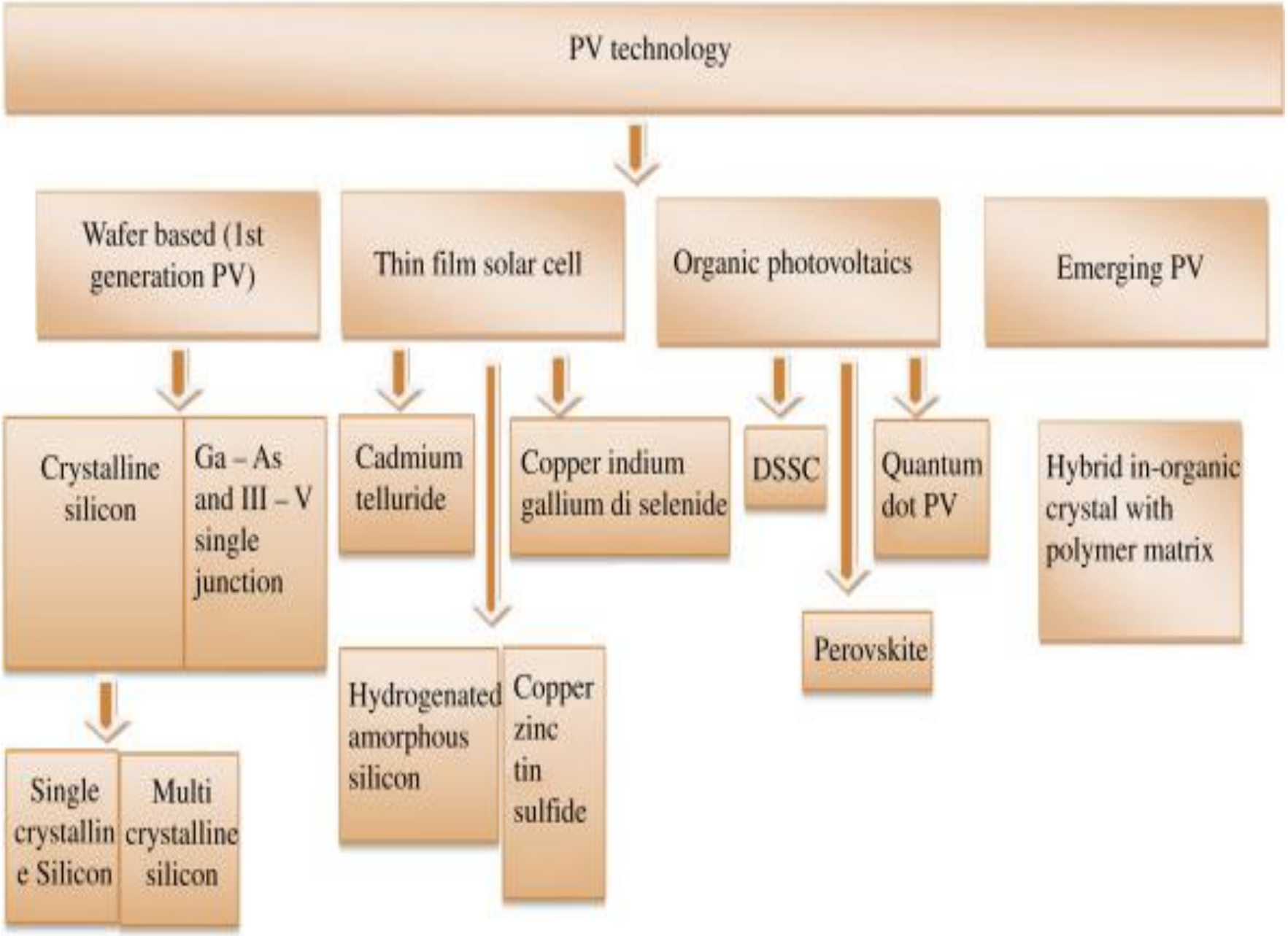
	1G	2G	3G
Technology	Wafer-based (single junction, mono- or polycrystalline silicon)	Thin film (CdTe, CIGS, a-Si)	Organic (Polymer, Small molecule), DSSC
Advantages	High quality, low-defect, high efficiency	High material utilisation, lower cost	Non-toxic, abundant, low-cost, short payback, transparent
Challenges	High consumption of active material (Si)	Scarcity or toxicity of some materials	Optimizing lifetime-efficiency-cost trade-off

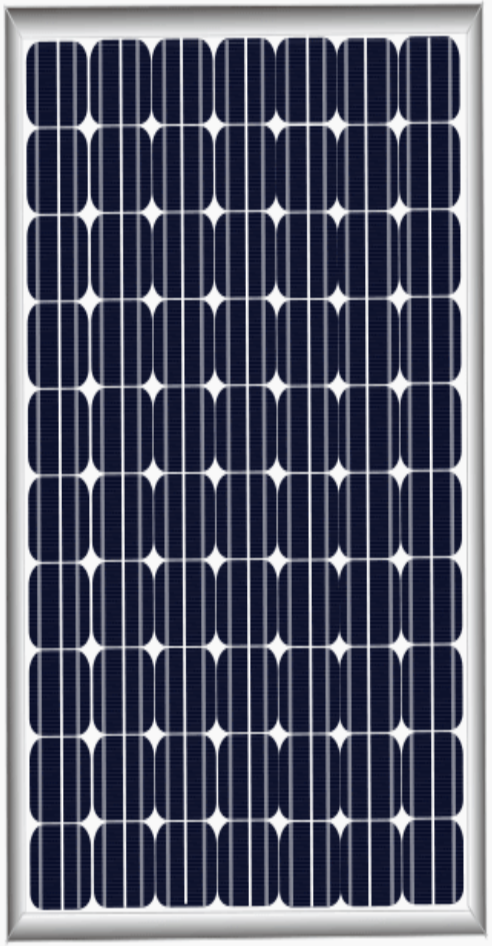

First Generation
Low Efficiency
High Cost
Second Generation
High Efficiency
Low Cost
Third Generation
Most Efficient
Cost –(depends)
Emerging 3rd.Generation
Rare Use – Under progress – high cost
Generations @
Crystallinity of Material
Band gap
Absorption Capacity
Manufacturing Complexity

Material	Properties	Advantages	Disadvantages
Monocrystalline Silicon Cells (MSC)	made out of 1 crystal, high surface area, low electrical resistance, typical Voltage=.5 V, typical current= 3A	very efficient >20%	expensive process to make crystal structure
Polycrystalline	multiple grains, this disturbs light flow	less cost than MSC	less efficient than MSC
Gallium Arsenide (GaAs)	made of Gallium, used in space applications	most efficient 20-25%	very expensive
Amorphous Silicon	2nd Generation	very cheap	less efficient (4-8%) in field
Copper indium diselenide	2nd Generation	N/A	N/A
Copper Indium gallium diselenide	thin film, 2nd generation	Highest efficiency for thin film (~17% lab)	very expensive

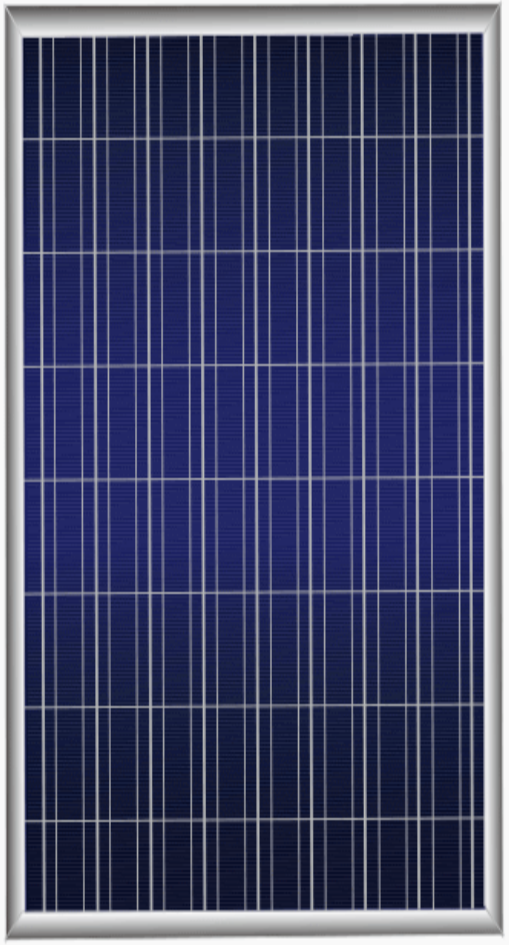
Comparison of Mono-Si, Poly-Si and Thin film Panels

Mono-Si Panels	Poly-Si Panels	Thin Film Panels
1. Most efficient with max. efficiency of 21%.	1. Less efficient with efficiency of 16% (max.)	1. Least efficient with max. efficiency of 12%.
2. Manufactured from single Si crystal.	2. Manufactured by fusing different crystals of Si.	2. Manufactured by depositing 1 or more layers of PV material on substrate.
3. Performance best at standard temperature.	3. Performance best at moderately high temperature.	3. Performance best at high temperatures.
4. Requires least area for a given power.	4. Requires less area for a given power.	4. Requires large area for a given power.
5. Large amount of Si hence, high embodied energy.	5. Large amount of Si hence, high Embodied energy.	4. Low amount of Si used hence, low embodied energy.
6. Performance degrades in low-sunlight conditions.	6. Performance degrades in low-sunlight conditions.	5. Performance less affected by low-sunlight conditions.
7. Cost/watt: 1.589 USD	7. 1.418 USD	7. 0.67 USD
8. Largest Manufacturer: Sunpower (USA)	8. Suntech (China)	8. First Solar (USA)

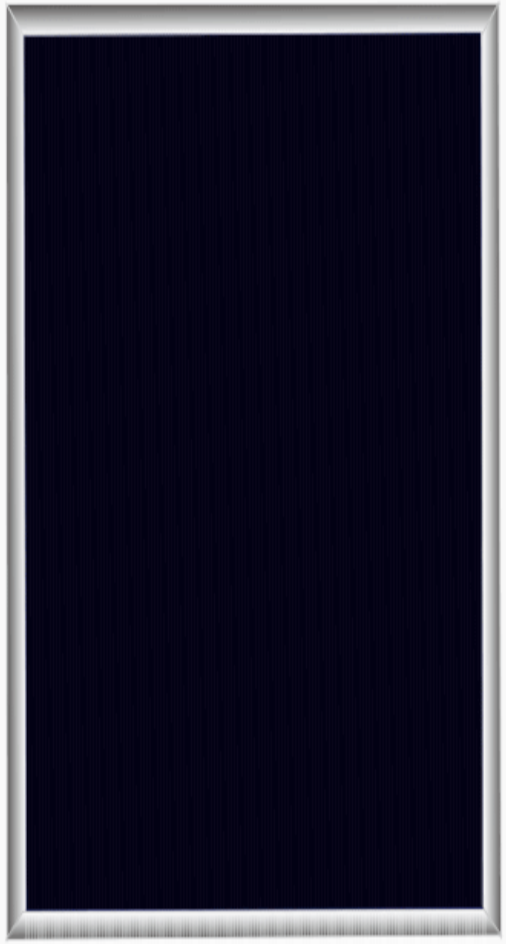

PV technology
Wafer based (1st generation PV)
Thin film solar cell
Organic photovoltaics
Emerging PV
Crystalline silicon
Ga – As and III – V single junction
Cadmium telluride
Copper indium gallium di selenide
DSSC
Quantum dot PV
Hybrid in-organic crystal with polymer matrix
Perovskite
Hydrogenated amorphous silicon
Copper zinc tin sulfide
Single crystalline Silicon
Multi crystalline silicon



Monocrystalline Solar Panel

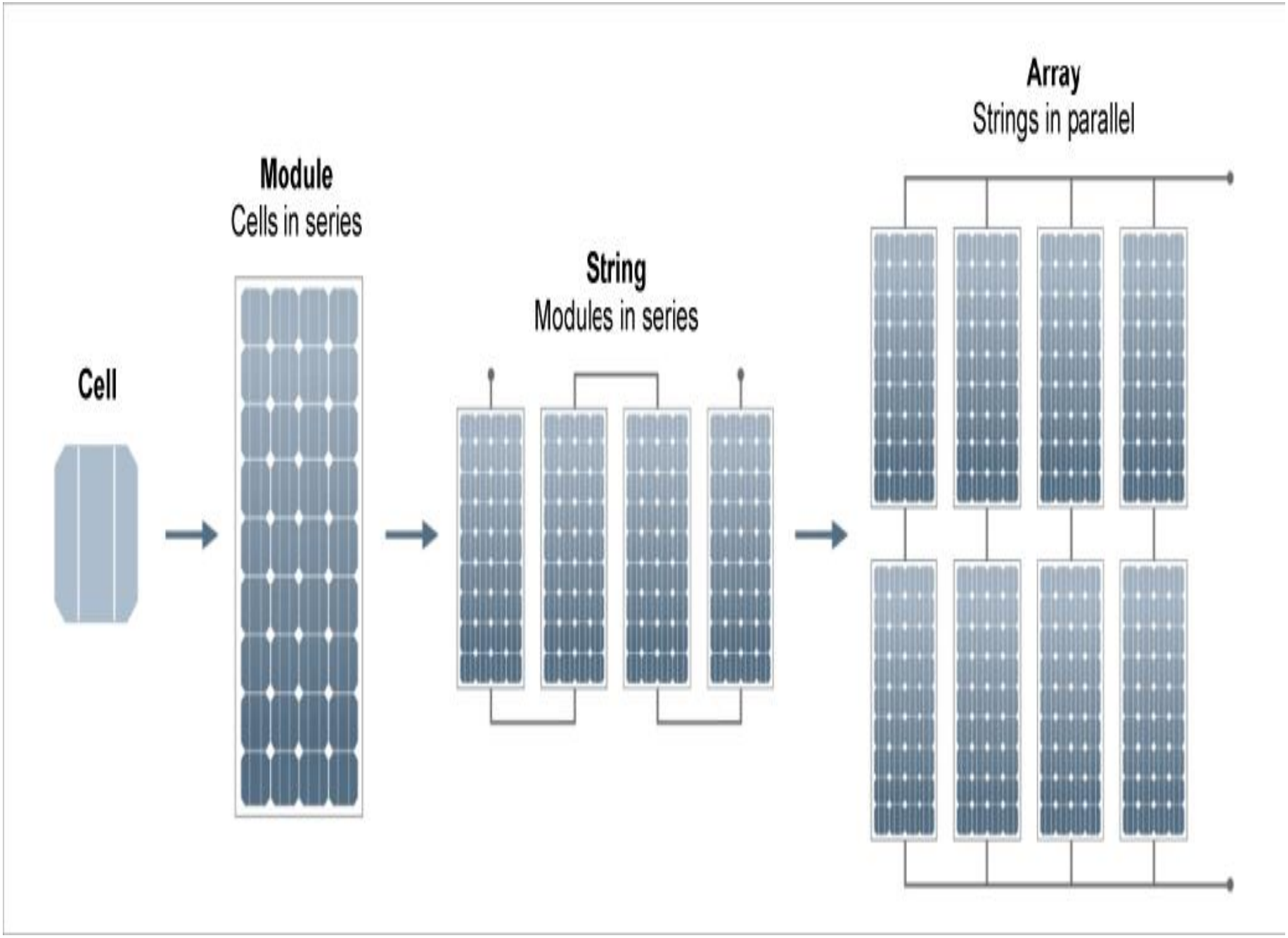


Polycrystalline Solar Panel



Thin-Film Solar Panel

© Solar Magazine

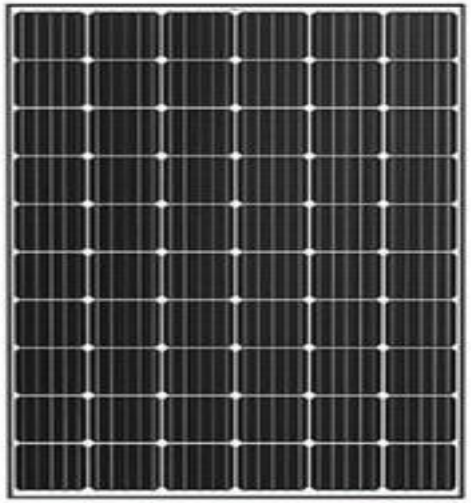

Cell
Module
Cells in series
String
Modules in series
Array
Strings in parallel

Average Efficiency of Different Solar Panels

CLEAN ENERGY REVIEWS

60 cell - Poly

16 - 17%

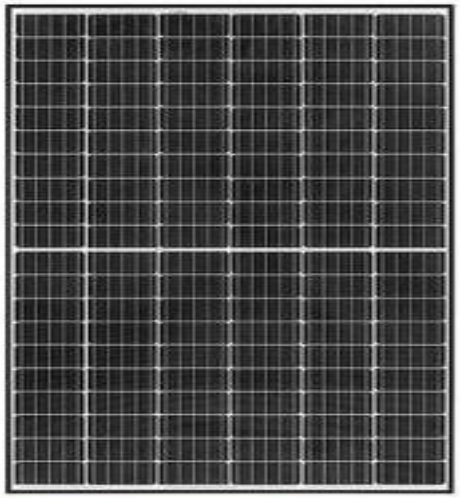


60 cell - Mono PERC

17 - 19%

Shingled cells - Mono

17 - 19%

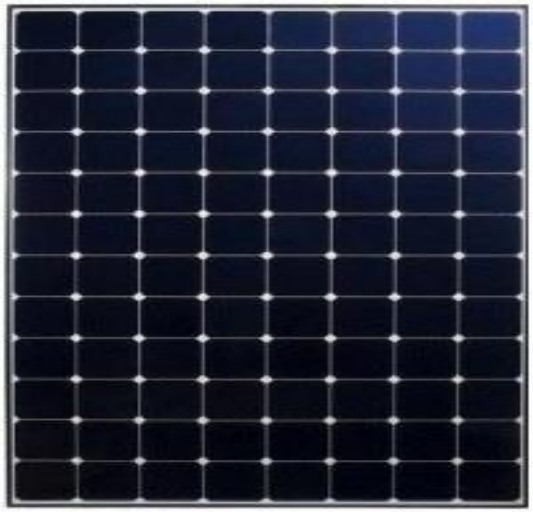


120 cell - Half-cut mono

18 - 20%

60 cell - Multi busbar

18 - 20%



60 or 96 cell - IBC

20 - 22%

www.cleanenergyreviews.info

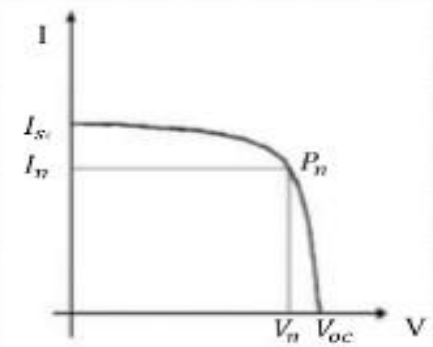
The different types of solar cells.

- Amorphous Silicon solar cell (a-Si)
- Biohybrid solar cell
- Cadmium telluride solar cell (CdTe)
- Concentrated PV cell (CVP and HCVP)
- Copper indium gallium selenide solar cells (CI(G)S)
- Crystalline silicon solar cell (c-Si)
- Float-zone silicon
- Dye-sensitized solar cell (DSSC)
- Gallium arsenide germanium solar cell (GaAs)
- Hybrid solar cell
- Luminescent solar concentrator cell (LSC)
- Micromorph (tandem-cell using a-Si/µc-Si)

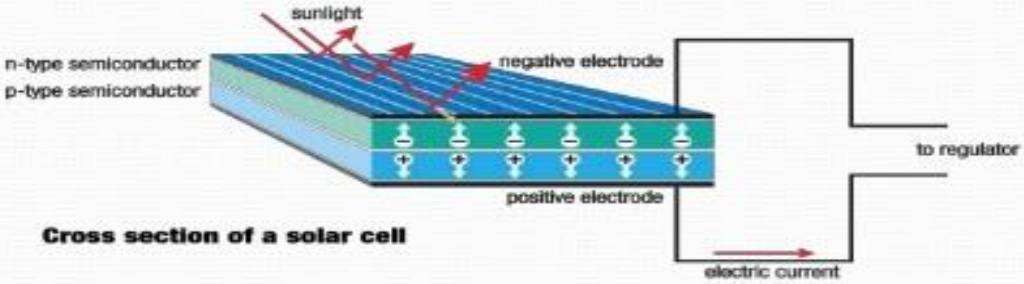
- Monocrystalline solar cell (mono-Si)
- Multi-junction solar cell (MJ)
- Nanocrystal solar cell
- Organic solar cell (OPV)
- Perovskite solar cell
- Photoelectrochemical cell (PEC)
- Plasmonic solar cell
- Polycrystalline solar cell (multi-Si)
- Quantum dot solar cell
- Solid-state solar cell
- Thin-film solar cell (TFSC)
- Wafer solar cell, or *wafer-based solar cell crystalline*
- Non concentrated hetrogeneos PV cell

Solar Cell Characteristics

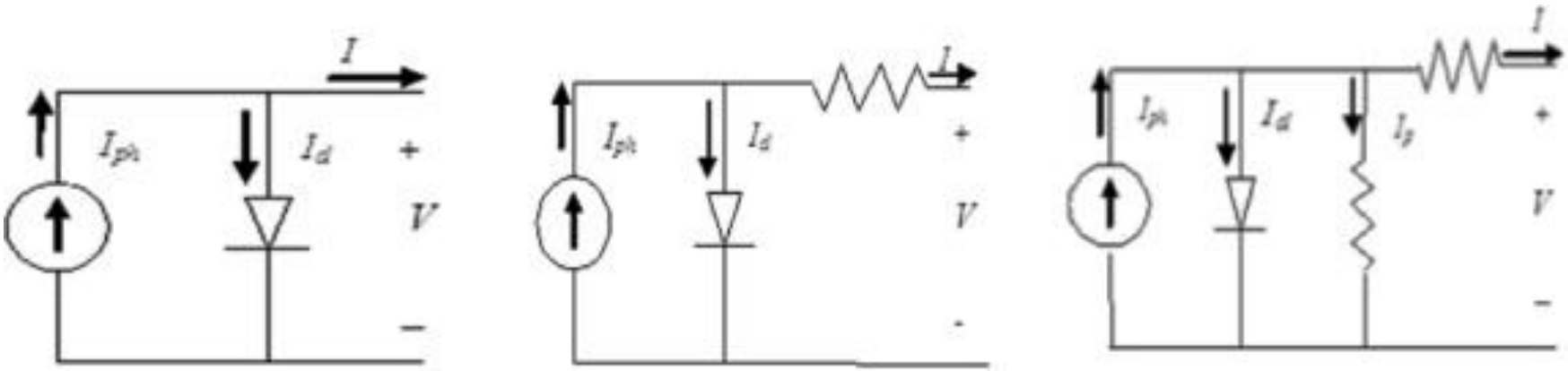
- Short circuit current (I_{sc})
- Open circuit voltage (V_{oc})
- Fill factor (FF)
- Efficiency (η)



I-V curve of solar cell and its parameter



Cross section of a solar cell



a) Ideal single diode model b) Practical model with Rs c) Practical model with Rs and Rp

$$I = I_{PV,cell} - I_{0,cell}\left[\exp\left(\frac{qV}{akT}\right) - 1\right]$$

$$E = \frac{hc}{\lambda}$$

If E is in eV and λ is in μm:

Where

I_{pv} - photo current

I_o - cell saturation of dark current

V_t - Thermal voltage which is $V_t = kT / q$,

q - 1.6×10^{-19} C charge of an electron.

T - the cell's working temperature

a - an ideality factor

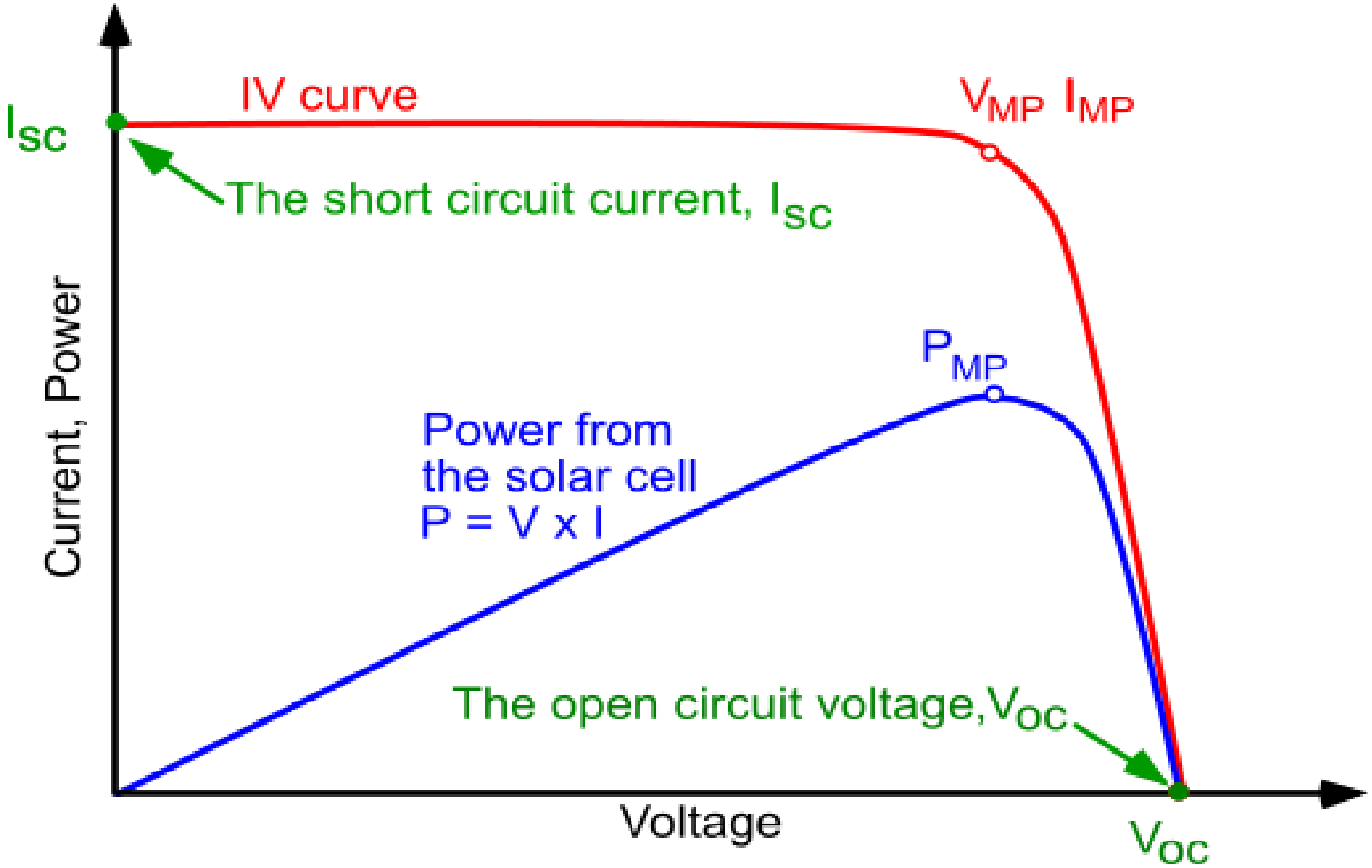
R_p - Shunt resistance

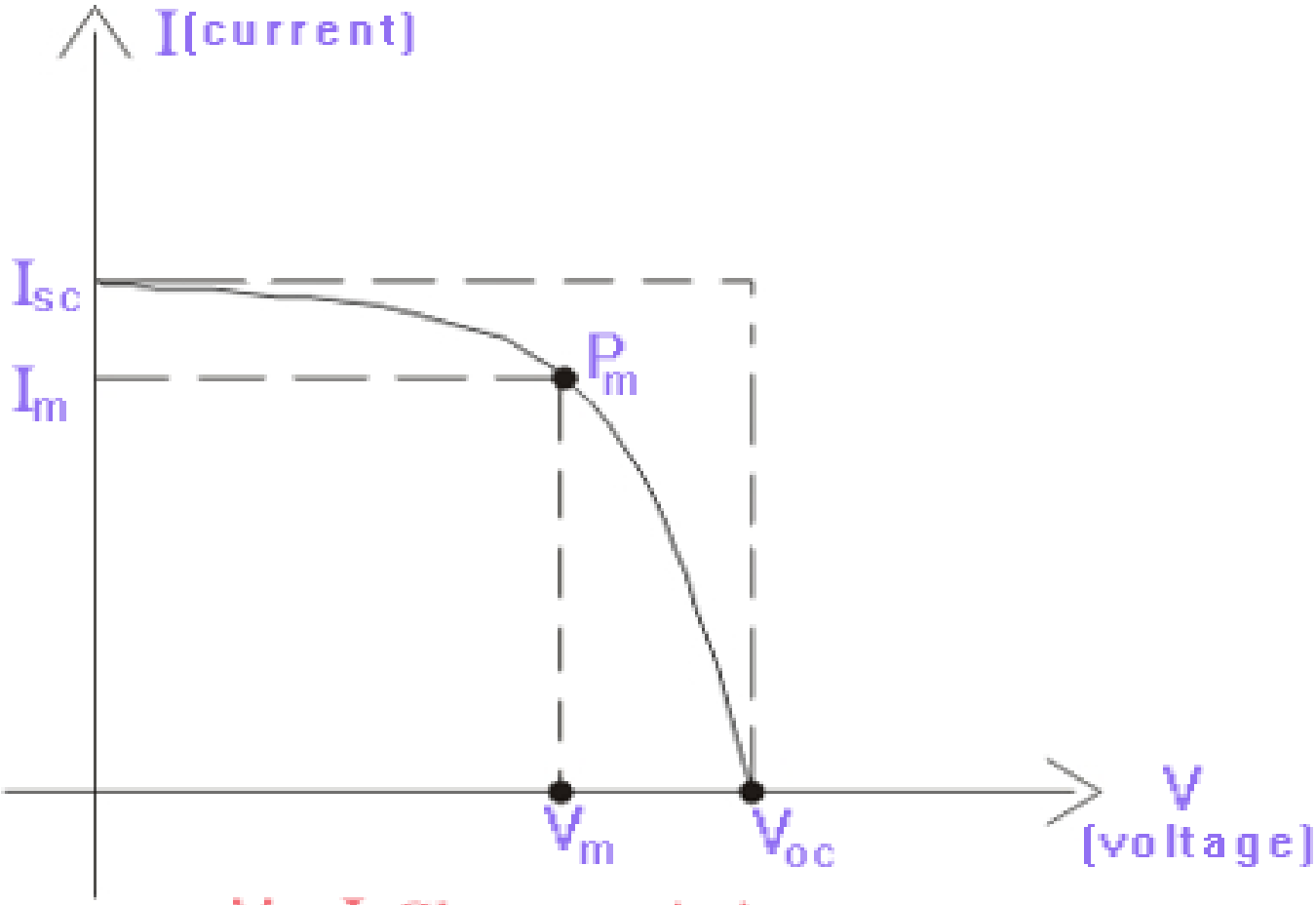
R_s - Series resistance

- **V_{OC}** = open-circuit voltage: – This is the maximum voltage that the array provides when the terminals are not connected to any load (an open circuit condition).
- This value is much higher than Vmp which relates to the operation of the PV array which is fixed by the load.
- This value depends upon the number of PV panels connected together in series.

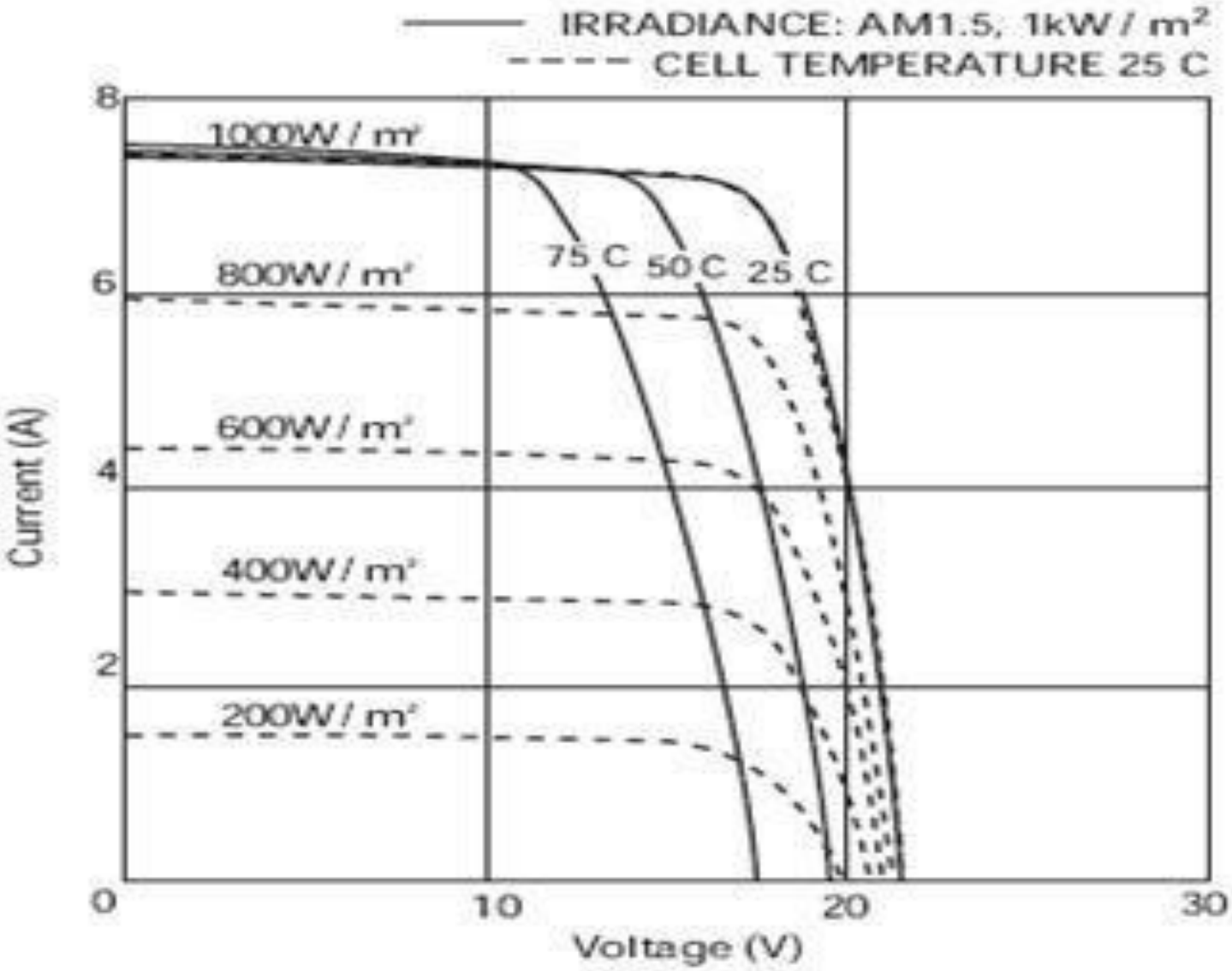
- **I_{SC}** = short-circuit current – The maximum current provided by the PV array when the output connectors are shorted together (a short circuit condition).
- This value is much higher than Imp which relates to the normal operating circuit current.

- **MPP** = maximum power point – This relates to the point where the power supplied by the array that is connected to the load (batteries, inverters) is at its maximum value, where MPP = Imp x Vmp.
- The maximum power point of a photovoltaic array is measured in Watts (W) or peak Watts (Wp).
- **FF** = fill factor – The fill factor is the relationship between the maximum power that the array can actually provide under normal operating conditions and the product of the open-circuit voltage multiplied by the short-circuit current, (V_{OC} x I_{SC}) This fill factor value gives an idea of the quality of the array and the closer the fill factor is to 1 (unity), the more power the array can provide. Typical values are between 0.7 and 0.8.
- **%eff** = percent efficiency – The efficiency of a photovoltaic array is the ratio between the maximum electrical power that the array can produce compared to the amount of solar irradiance hitting the array.
- The efficiency of a typical solar array is normally low at around 10-12%, depending on the type of cells (monocrystalline, polycrystalline, amorphous or thin film) being used.


IV curve
V_{MP} I_{MP}
I_{SC}
The short circuit current, I_{SC}
P_{MP}
Power from
the solar cell
P = V x I
The open circuit voltage, V_{OC}
Current, Power
Voltage
V_{OC}



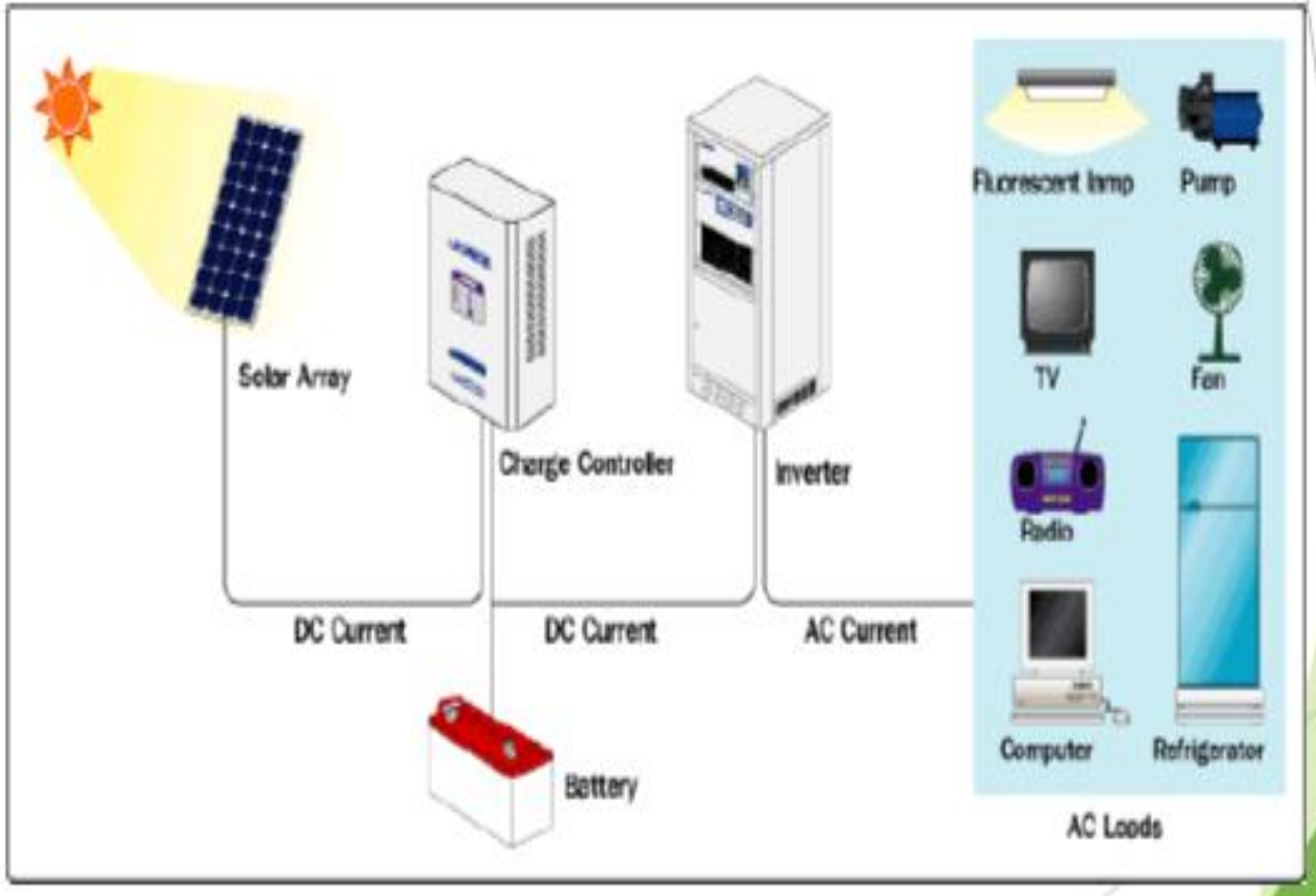
V - I Characteristics


IRRADIANCE: AM1.5, 1kW / m²
CELL TEMPERATURE 25 C
1000W / m²
800W / m²
600W / m²
400W / m²
200W / m²
75 C
50 C
25 C
Current (A)
Voltage (V)
0
2
4
6
8
10
20
30

Types of Solar Photovoltaic System

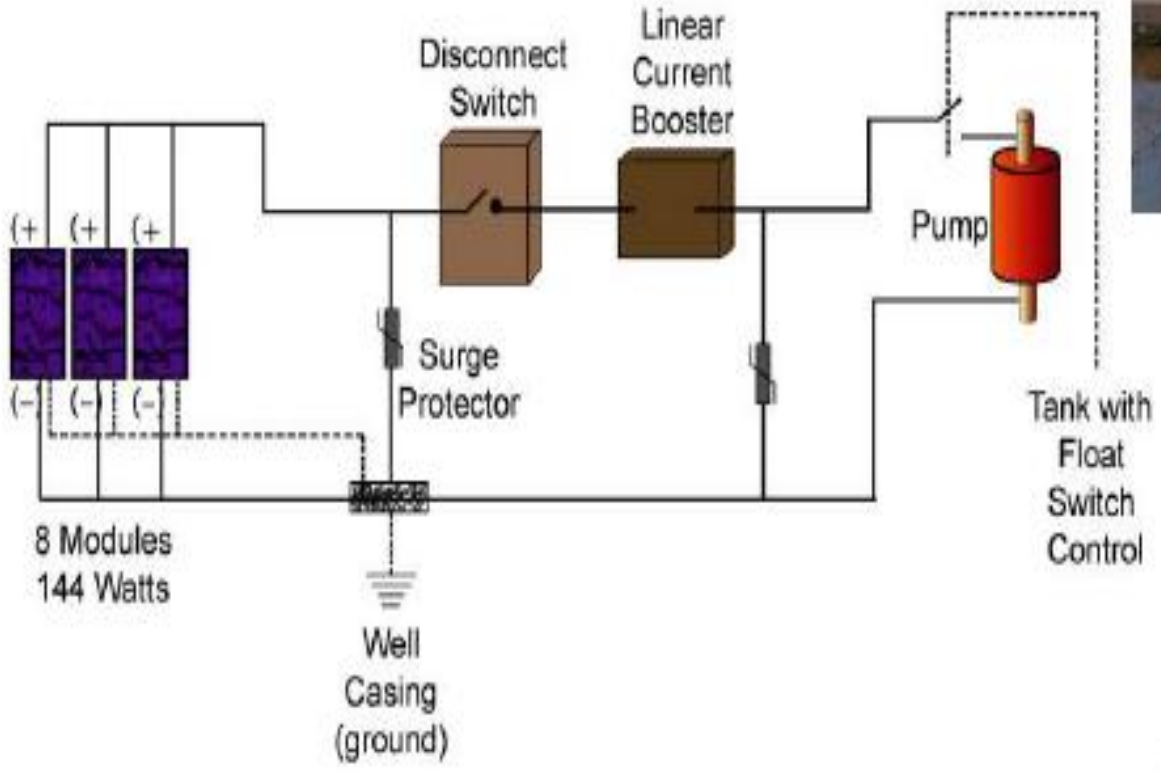
- **Stand-alone systems** - those systems which use photovoltaics technology only, and are not connected to a utility grid.
- **Hybrid systems** - those systems which use photovoltaics and some other form of energy, such as diesel generation or wind.
- **Grid-tied systems** - those systems which are connected to a utility grid.

Stand Alone PV System

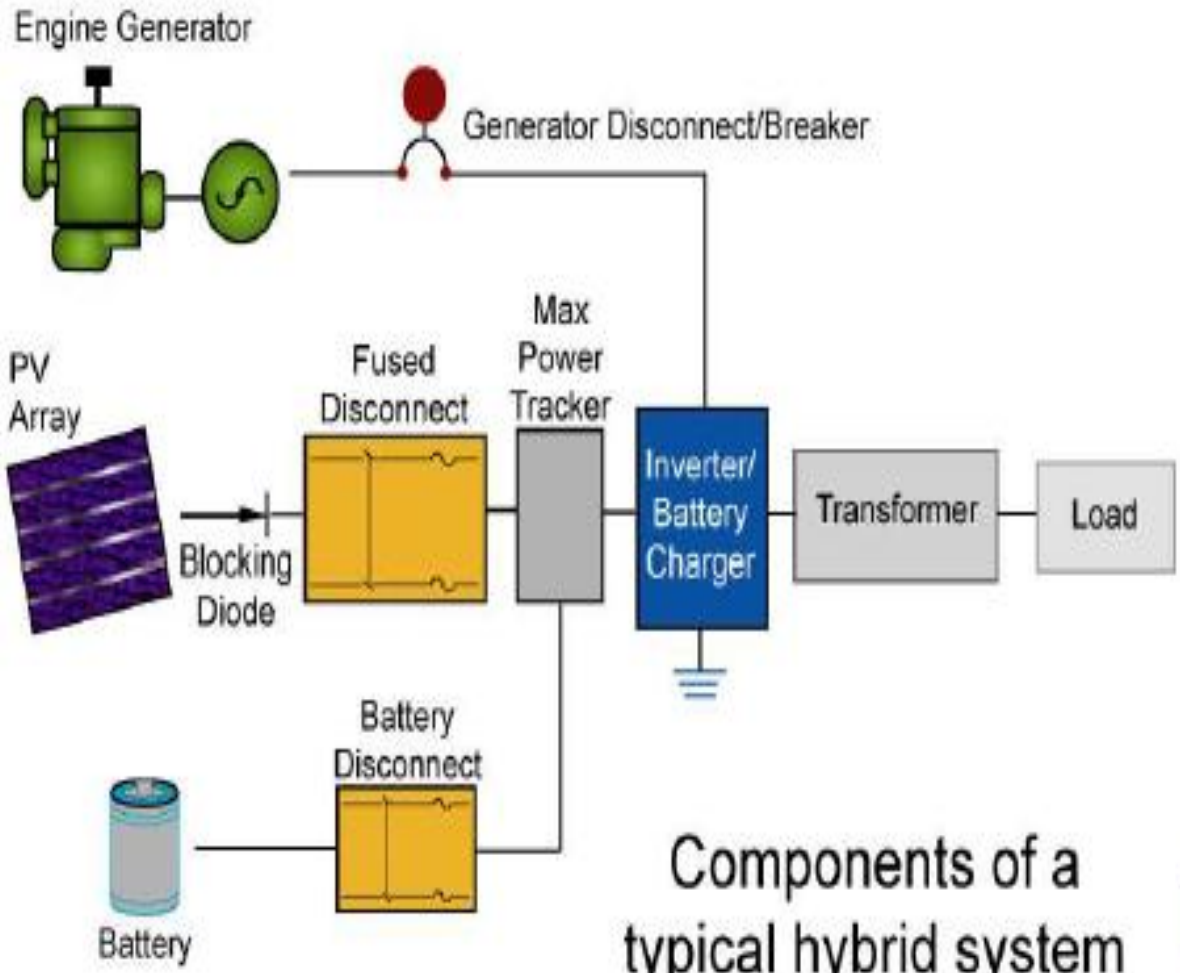


Stand Alone PV System

Water pumping system

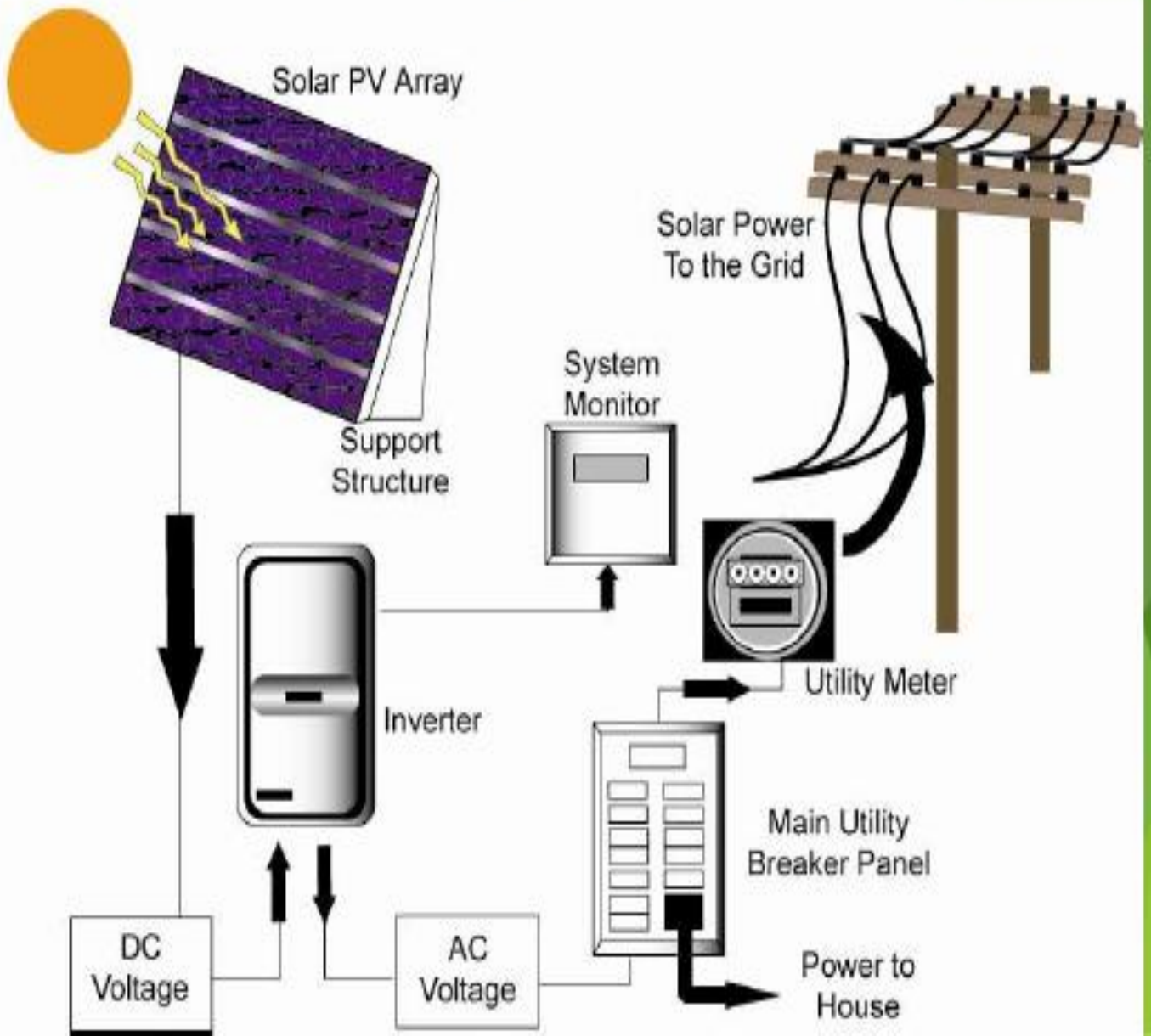
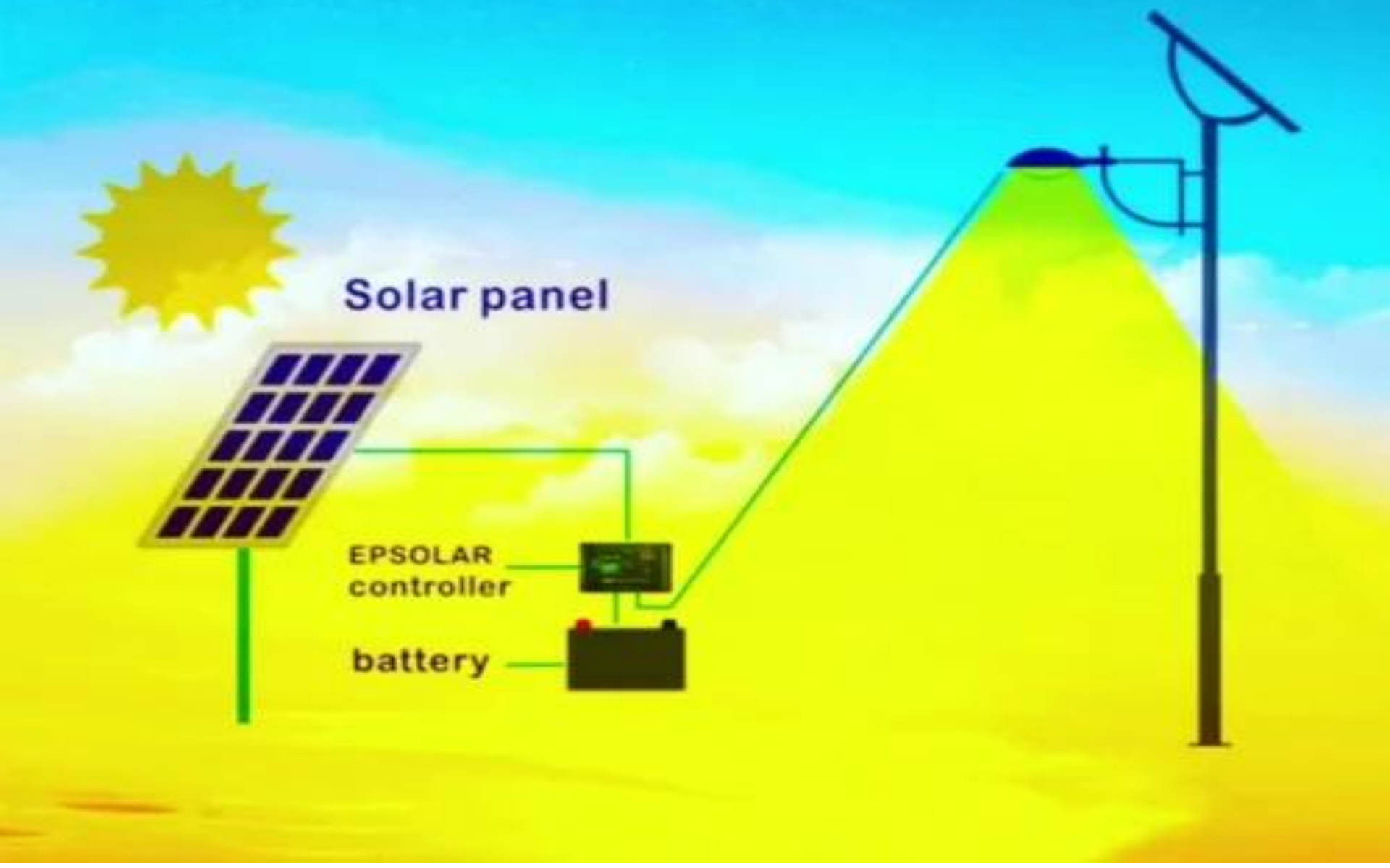


Hybrid PV System



Components of a typical hybrid system

Grid-Tied PV System



Solar LED Lights